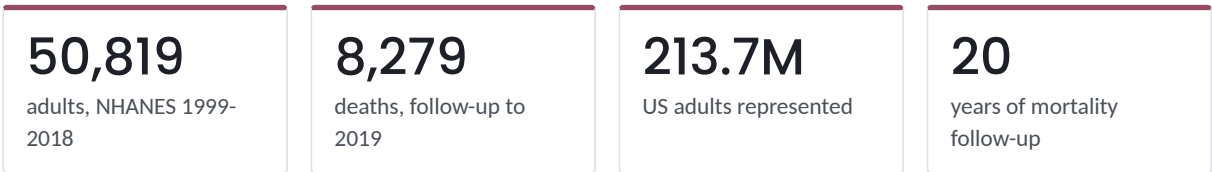
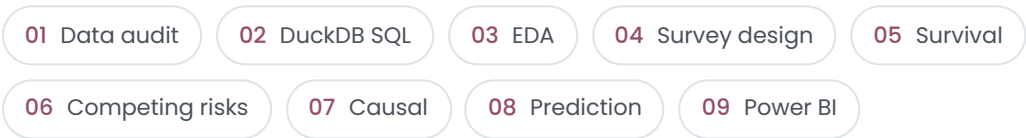


Cardiometabolic Risk and Mortality in US Adults

Survey-weighted survival analysis of NHANES 1999-2018 linked to the National Death Index, with competing risks, causal inference and a 10-year risk model



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Python, DuckDB SQL, R, Power BI

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Executive summary

This project follows 50,819 US adults from the National Health and Nutrition Examination Survey (NHANES, 1999 to 2018) until the end of 2019, using the NCHS link to the National Death Index. During a median of 9 years, 8,279 of them died. The aim is to measure how diabetes, prediabetes, hypertension, obesity and chronic kidney disease (CKD) relate to death from any cause and from specific causes, and how well 10-year risk of death can be predicted from one exam.

Python and DuckDB SQL handle the download, cleaning and descriptive work. All statistics are done in R with the full survey design (weights, strata and PSUs), so the estimates describe the 213.7 million US adults the sample represents. Missing values are filled with 20 multiple imputations, and the results feed an 8-page Power BI dashboard.

11.8%

diabetes, age-standardized, of which 23.4% undiagnosed

1.44

hazard ratio for death with diabetes, fully adjusted

14.7%

of deaths attributable to CKD, the largest share

0.872

C-index of the 10-year risk model on later cycles

Main findings

- 1. Prediabetes carried no excess risk of death once age was accounted for.** Crude, its death rate was nearly twice that of normal glucose, but people with prediabetes were 11 years older on average. Fully adjusted, the hazard ratio was 0.99 (95% CI 0.92 to 1.06), and it stayed close to 1 for every cause of death and every sensitivity analysis.
- 2. Diabetes and CKD were the strongest cardiometabolic risk factors.** Adjusted for age, sociodemographics, smoking and the other conditions, diabetes raised the rate of death by 44% (HR 1.44, 1.34 to 1.55) and CKD by 59% (HR 1.59, 1.50 to 1.68). Counting its effect through blood pressure and kidneys, the total effect of diabetes from inverse probability weighting was HR 1.83 (1.58 to 2.13), or 6.3 extra deaths per 100 people over 10 years.
- 3. The excess risk with diabetes was in CVD and other causes, not cancer.** The cause-specific hazard ratio for diabetes was 1.50 for CVD, 1.62 for other causes and 1.03 for cancer. Diabetes appeared on the death certificate of only 34.5% of people who died with diagnosed diabetes.
- 4. CKD accounted for the largest share of deaths.** 14.7% of all deaths were attributable to CKD, ahead of current smoking (13.1%), hypertension (11.8%) and diabetes (8.3%). For CVD deaths, hypertension and CKD each accounted for over a fifth.
- 5. Diabetes and obesity rose, and undiagnosed diabetes did not fall.** Age-standardized diabetes went from 9.0% in 1999-2000 to 14.4% in 2017-2018 and obesity from 30.3% to 42.5%. About a quarter of diabetes was undiagnosed in every cycle.
- 6. Ten-year risk of death can be predicted well from one exam.** A Cox model with demographics, smoking, conditions and routine lab values, built on 1999-2006 and tested on 2007-2010, reached a C-index of 0.872 with close calibration (observed 10.4%, predicted 10.2%).

What it means

For screening and prevention, CKD deserves more attention than it usually gets. It is common, often silent, and in this data it was linked to more deaths than diabetes. Finding the quarter of diabetes that stays undiagnosed is still an open problem after 20 years. A prediabetes result by itself is a weak signal of risk of death, so it is better treated as one input among several, as in the risk model. These are associations from observational data measured once at baseline (Section 11).

1 Introduction

What the project asks, what data it uses, and how the work is split between Python, SQL, R and Power BI.

1.1 Background

Diabetes, high blood pressure, obesity and chronic kidney disease are common in US adults, and they tend to come together in the same people. Most studies of their effect on death use either self-reported conditions or patients who are already in care. That misses one group completely: people who have the condition and do not know it.

The National Health and Nutrition Examination Survey (NHANES) does not have this problem. Every participant gets a physical exam and blood and urine tests in a mobile examination centre, so diabetes and kidney disease are measured directly, not only asked about. The National Center for Health Statistics (NCHS) links adult participants to the National Death Index, which gives the date and underlying cause of death for anyone who died before the end of 2019. Together these two sources support a proper cohort study of the US adult population, with up to 20 years of follow-up.

NHANES is a complex survey. People are not sampled at random. Some groups are oversampled on purpose, and participants are clustered in sampled counties. Any estimate that is meant to describe US adults has to use the survey weights, strata and primary sampling units, and that shapes almost every method choice in this report.

1.2 Questions

1. **How common are these conditions, and did that change between 1999 and 2018?** Weighted and age-standardized prevalence, with trend tests (Section 5).
2. **How much higher is the risk of death for each condition, after adjusting for age and everything else?** Survey-weighted Kaplan-Meier curves, Cox models, dose-response splines and sensitivity analyses (Section 6).
3. **What do people with each condition die of?** Competing risks: cause-specific and Fine-Gray models, and cumulative incidence (Section 7).
4. **How many deaths could be linked to each risk factor?** Inverse probability weighting, E-values and population attributable fractions (Section 8).
5. **Can 10-year risk of death be predicted from a single exam?** Three models with temporal validation, calibration and decision curves (Section 9).

1.3 Data

Table 1.1 Data used in this project.

Item	Detail
Survey	NHANES continuous, 10 two-year cycles, 1999-2000 to 2017-2018 (CDC, NCHS)
Components	14 per cycle: demographics, body measures, blood pressure exam, HbA1c, fasting glucose, total and HDL cholesterol, triglycerides, urine albumin and creatinine, biochemistry, and diabetes, blood pressure, medical conditions and smoking questionnaires
Mortality	NCHS public-use Linked Mortality Files, follow-up to 31 December 2019, with underlying cause of death
Raw size	101,316 participants of all ages, 140 component files and 10 mortality files
Analytic cohort	50,819 adults, 8,279 deaths, 479,971 person-years (Section 2)
Population represented	213.7 million US adults aged 20 and over (weighted)

1.4 Workflow

The work runs as nine numbered steps. Python notebooks in Google Colab download the data, audit and clean it, answer descriptive questions in DuckDB SQL and explore the data. R scripts in RStudio then do all the statistics, because the R `survey` package handles weighted survival models and multiple imputation together. The last R script writes the tables for the Power BI dashboard, so the dashboard only displays results and does not recompute them.

Table 1.2 Workflow steps and what each one produces.

Step	Tool	Name	What it does
01	Python	Download	NHANES components and mortality files for 10 cycles
02	Python	Data audit	Harmonize, derive conditions, cohort rules, missingness
03	Python, SQL	DuckDB SQL	11 descriptive questions, crude and unweighted
04	Python	EDA	Skew, correlation, missing data, risk shapes, age distortion
05	R	Survey design and imputation	Survey design, multiple imputation, weighted prevalence
06	R	Survival and Cox	Death rates, Kaplan-Meier, Cox models, splines, sensitivity
07	R	Competing risks	Cause of death, cumulative incidence, Fine-Gray
08	R	Causal and attributable	IPW, balance, E-values, attributable fractions
09	R	Risk prediction and export	10-year risk models, validation, Power BI tables
	Power BI	Dashboard	8 report pages, drill-through and tooltip pages

1.5 Reading the numbers

Three kinds of numbers appear in this report, and they are not interchangeable.

- **Crude, unweighted** (Sections 3 and 4). These describe the people in the sample. NHANES oversamples older adults, Black and Hispanic adults and low-income groups, so these numbers are not US estimates. They are used to check the data and to plan the models.
- **Weighted** (Sections 5 to 9). These use the survey design and describe US adults aged 20 and over. They come with 95% confidence intervals that account for strata and clustering.
- **Age-standardized**. Rates re-weighted to the 2000 US standard population, so that groups with different age mixes can be compared fairly. Age is by far the strongest driver of death in this data, so this matters a lot (Section 4.5).

Death rates are per 1,000 person-years: deaths divided by the total years of follow-up, which handles people being followed for different lengths of time. A hazard ratio (HR) of 1.44 means the rate of death at any point in follow-up is 44% higher than in the reference group, holding the other variables in the model fixed.

2 Data preparation and cohort

Notebooks 01 and 02. Downloading 14 NHANES components for 10 cycles, linking mortality, and building one analysis-ready row per adult.

2.1 Download

Notebook 01 downloads each component as a SAS transport (`.xpt`) file from the CDC site, for every cycle. The file names are not consistent. Before 2005-2006 several lab components used different names (HbA1c was `LAB10` , `L10_B` and `L10_C` before it became `GHB_D`), and HDL cholesterol sat inside the total cholesterol file. A small resolver tries the known candidate names for each cycle and logs which one worked. All 140 component files resolved, and the manifest has no missing rows.

Code 2.1 Resolving the right file name per cycle before downloading.

01_download_nhanes_mortality.ipynb, cell 6

```
def resolve_and_fetch(stem, cycle_key):
    year, suffix = CYCLES[cycle_key]["year"], CYCLES[cycle_key]["suffix"]
    candidates = COMPONENTS[stem]["legacy"] if year < 2005 else [stem]
    for name in candidates:
        url = f"https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/{year}/DataFiles/{name}{suffix}.xpt"
        r = requests.get(url, headers=HEADERS, timeout=30)
        if r.ok and r.content[:2] != b"<!":
            return url, r.content
    return None, None
```

The mortality files are fixed-width text. Column positions come from the CDC sample read-in program. The ten files hold 101,316 records, one per NHANES participant, with eligibility, vital status, leading underlying cause of death and months of follow-up from the interview and from the exam.

2.2 Harmonizing variables

Some variable names changed between cycles. HDL is `LBDHDL` in 1999-2002, `LBXHDD` in 2003-2004 and `LBDHDD` from 2005. Serum creatinine is `LBDSCR` in 2001-2002, and diabetes pills are `DID070` in 2005-2008. Each analysis variable gets a list of candidate names, and the first non-missing one wins. A coverage check by cycle then looks for any variable at 0% in a cycle, which would mean a missed rename. The only zeros were the 4-year weights, which exist only in the first two cycles, as expected.

One quirk needed a fix before anything else. `pandas.read_sas` reads a stored zero as about $5.4e-79$. That is harmless for most columns but breaks any rule that tests for zero, such as a diastolic reading of 0, which NHANES uses for "not measurable". Values below $1e-70$ are set to exactly 0.

2.3 Defining the conditions

Table 2.1 How each condition and outcome is defined.

Variable	Definition
Diabetes	Told by a doctor, on insulin or pills, HbA1c 6.5% or higher, or fasting glucose 126 mg/dL or higher (ADA criteria plus self-report)
Undiagnosed diabetes	Meets the lab criteria but never told, not on insulin or pills
Prediabetes	No diabetes, and HbA1c 5.7% to 6.4%, fasting glucose 100 to 125 mg/dL, or told "borderline"
Hypertension	Mean systolic 140+ or diastolic 90+ mmHg, or on blood pressure medication (JNC7). ACC/AHA 130/80 kept as a second flag
Obesity, BMI groups	BMI under 18.5 underweight, 18.5 to 24.9 normal, 25 to 29.9 overweight, 30+ obese
eGFR	CKD-EPI 2021 race-free equation on calibrated serum creatinine
Chronic kidney disease	eGFR below 60 mL/min/1.73m ² or urine albumin-to-creatinine ratio (ACR) 30 mg/g or higher
Prior CVD	Self-reported heart failure, coronary heart disease, angina, heart attack or stroke
Smoking	Never (under 100 cigarettes in life), former, current
Cause of death	NCHS leading cause recode: heart disease and stroke as CVD, cancer, all others as other

NHANES questionnaire codes 7 (refused) and 9 (don't know) become missing. Several questions are only asked after a "yes". Diabetes pills, for example, are only asked of people told they have diabetes. A missing answer there counts as "no" when building a composite condition, not as unknown, otherwise almost everyone would end up with a missing diabetes status.

Code 2.2 Deriving diabetes, prediabetes and hypertension. A condition is missing only when the key measurement is missing.

02_data_audit.ipynb, cell 10

```
def flag(conds, known):
    hit = pd.concat(conds, axis=1).any(axis=1)
    return pd.Series(np.where(hit, 1.0, np.where(known, 0.0, np.nan)), index=d.index)
d["diabetes"] = flag([d.dm_told.eq(1), d.dm_insulin.eq(1), d.dm_pills.eq(1), d.hba1c.ge(6.5), d.fpg.ge(126)],
d.hba1c.notna() & d.dm_told.notna())
d["prediabetes"] = flag([d.hba1c.between(5.7, 6.4), d.fpg.between(100, 125), d.dm_borderline.eq(1)],
d.hba1c.notna()).where(d.diabetes.ne(1), 0.0)
d["glycemic_status"] = np.select([d.diabetes.eq(1), d.prediabetes.eq(1), d.diabetes.eq(0) & d.prediabetes.eq(0)],
["diabetes", "prediabetes", "normal"], None)

d["hypertension"] = flag([d.sbp.ge(140), d.dbp.ge(90), d.htn_med.eq(1)], d.sbp.notna())
d["hypertension_acc"] = flag([d.sbp.ge(130), d.dbp.ge(80), d.htn_med.eq(1)], d.sbp.notna())
```

Serum creatinine in 1999-2000 and 2005-2006 was measured on assays that drifted from the standard reference. The published CDC corrections are applied before eGFR is calculated.

Code 2.3 Creatinine calibration, CKD-EPI 2021 eGFR and the CKD flag.

02_data_audit.ipynb, cell 12

```
d["scr_std"] = np.select([d.cycle.eq("1999-2000"), d.cycle.eq("2005-2006")], [1.013 * d.scr + 0.147, 0.978 * d.scr -
0.016], d.scr)
female = d.sex.eq(2)
r = d.scr_std / np.where(female, 0.7, 0.9)
d["egfr"] = 142 * np.minimum(r, 1) ** np.where(female, -0.241, -0.302) * np.maximum(r, 1) ** -1.200 * 0.9938 ** d.age *
np.where(female, 1.012, 1)
d["ckd"] = flag([d.egfr.lt(60), d.albuminuria.eq(1)], d.egfr.notna() | d.acr.notna())
```

2.4 Cohort

The cohort is adults aged 20 and over who completed the exam, were eligible for mortality linkage and were not pregnant. Follow-up starts at the exam, not the interview, because labs and blood pressure are measured at the exam. Every exclusion is counted.

Code 2.4 Cohort rules, applied in order, with the count after each step.

02_data_audit.ipynb, cell 17

```

RULES = [
    ("Age 20+", lambda x: x.age >= 20),
    ("MEC examined", lambda x: x.exam_status == 2),
    ("Eligible for mortality linkage", lambda x: x.eligstat == 1),
    ("Not pregnant", lambda x: x.pregnant.ne(1)),
    ("Follow-up time and status available", lambda x: x.time_months.notna() & x.death.notna()),
]
cohort, funnel = d, [{"All NHANES participants 1999-2018", len(d)}]
for step, rule in RULES:
    cohort = cohort[rule(cohort)]
    funnel.append((step, len(cohort)))

```

Table 2.2 Cohort flow.

Step	Remaining	Excluded
All NHANES participants 1999-2018	101,316	0
Age 20+	55,081	46,235
MEC examined	52,398	2,683
Eligible for mortality linkage	52,287	111
Not pregnant	50,819	1,468
Follow-up time and status available	50,819	0

The final cohort has **50,819 adults** and **8,279 deaths** (16.3%) over 479,971 person-years. Median follow-up is 9.0 years and the longest is 20.8 years. Of the deaths, 2,593 were from CVD, 1,837 from cancer, 3,848 from other causes, and one had no cause recorded; it is counted with other causes in the R scripts. Only 111 examined adults were not eligible for linkage. They were slightly younger (mean age 46.7 vs 49.7) with similar diabetes prevalence, so leaving them out changes little.

2.5 Survey weights

The ten cycles are combined into one 20-year sample following the CDC rule. For 1999-2002 the 4-year exam weight is multiplied by 2/10. For every later cycle the 2-year exam weight is multiplied by 1/10. The combined weights add up to 213.7 million, the average US adult population over the period.

Code 2.5 Outcome, cause of death and combined 10-cycle weights.

02_data_audit.ipynb, cell 15

```

d["time_months"], d["death"] = d.permth_exm, d.mortstat
d["cause"] = np.select([d.mortstat.eq(0), d.ucod_leading.isin([1, 5]), d.ucod_leading.eq(2), d.ucod_leading.notna(),
d.mortstat.eq(1)], ["alive", "cvd", "cancer", "other", "unknown"], None)
early = d.cycle.isin(["1999-2000", "2001-2002"])
d["wt_mec"] = np.where(early, d.wt_mec4 * 2 / 10, d.wt_mec2 / 10)
d["wt_fast"] = np.where(early, d.wt_fast4 * 2 / 10, d.wt_fast2 / 10)

```

2.6 Missing data and range checks

Fasting glucose, triglycerides and LDL are measured only in the morning fasting subsample, so about 55% are missing by design. They are not imputed and not used in the main models. Among the other variables, income-to-poverty ratio is missing most often (9.0%), followed by eGFR (6.4%), HDL and total cholesterol (6.1% each) and waist (5.8%). Section 4.4 shows that these values are not missing at random, which is why the R work uses multiple imputation.

Table 2.3 Values outside plausible limits (only variables with at least one flag).

variable	low	high	below	above	min	max
bmi	12	80	0	7	12.0	130.2
dbp	20	150	24	0	3.3	138.7
hba1c	3	20	1	0	2.0	18.8
fpg	40	600	6	0	21.0	587.3
tc	50	700	0	3	59.0	813.0
hdl	5	200	0	1	6.0	226.0
egfr	1	200	0	1	2.0	242.0
acr	0	20,000	0	2	0.1	92,500.0

Only 45 values fall outside plausible limits, out of hundreds of thousands. They are CDC measurements taken under quality control, so they are kept. Their pull on the models is limited by fitting continuous measures as splines instead of straight lines. Age is top-coded by NHANES at 85 for 1999-2006 and at 80 from 2007, so only the 80+ age group is safe to use for the oldest people.

3 Descriptive questions in SQL

Notebook 03. Eleven questions a health department would ask, answered in DuckDB SQL straight from the cohort Parquet file. All numbers here are crude and unweighted.

DuckDB runs SQL directly on the Parquet file, without loading it into a database first. Each query sits in its own named variable, one clause per line, and each result is saved as a CSV. These numbers describe the 50,819 people in the sample and are not age-adjusted. They are a first look, and Section 4 shows how much of the pattern is really about age.

3.1 Death rate by glycemic status and burden

Table 3.1 Crude death rate by glycemic status (q02).

Glycemic status	Participants	Deaths	Deaths per 1,000 PY	Mean age
normal	24,885	2,808	10.76	43.9
prediabetes	15,175	2,555	19.42	54.9
diabetes	8,575	2,520	37.73	61.6

The crude death rate climbs from 10.76 per 1,000 person-years with normal glucose to 19.42 with prediabetes and 37.73 with diabetes. But mean age also climbs, from 43.9 to 61.6 years. Part of this gradient is age, not glucose.

Code 3.1 Counting diabetes, hypertension and obesity per person, then share of cohort and share of deaths with window functions (q03). 03_duckdb_sql.ipynb, cell 7

```
WITH scored AS (
  SELECT
    death,
    time_months,
    diabetes + hypertension + (bmi >= 30)::INT          AS n_conditions
  FROM cohort
  WHERE diabetes IS NOT NULL
    AND hypertension IS NOT NULL
    AND bmi IS NOT NULL
)
SELECT
  n_conditions::INT                                AS n_conditions,
  COUNT(*)                                          AS participants,
  ROUND(100 * COUNT(*) / SUM(COUNT(*)) OVER (), 1) AS pct_of_cohort,
  SUM(death)::INT                                  AS deaths,
  ROUND(100 * SUM(death) / SUM(SUM(death)) OVER (), 1) AS pct_of_deaths,
  ROUND(1000 * SUM(death) / (SUM(time_months) / 12), 2) AS deaths_per_1000_py
FROM scored
GROUP BY n_conditions
ORDER BY n_conditions
```

Table 3.2 Number of conditions (diabetes, hypertension, obesity) and death rate (q03).

Conditions	Participants	% of cohort	Deaths	% of deaths	Deaths per 1,000 PY
0	18,540	39.9	1,380	19.1	7.16
1	16,053	34.6	2,927	40.4	19.73
2	8,544	18.4	2,051	28.3	28.01
3	3,280	7.1	880	12.2	34.31

Each extra condition raises the crude death rate, from 7.16 with none to 34.31 with all three. People with one condition make up 34.6% of the cohort but 40.4% of deaths.

3.2 Riskiest profiles

Code 3.2 Every combination of four conditions, ranked by death rate, with profiles under 200 people left out (q04).

03_duckdb_sql.ipynb, cell 9

```
WITH profiles AS (
  SELECT
    CONCAT_WS(' + ',
      CASE WHEN diabetes = 1 THEN 'Diabetes' END,
      CASE WHEN hypertension = 1 THEN 'Hypertension' END,
      CASE WHEN bmi >= 30 THEN 'Obesity' END,
      CASE WHEN ckd = 1 THEN 'CKD' END) AS profile,
    death,
    time_months
  FROM cohort
  WHERE diabetes IS NOT NULL
    AND hypertension IS NOT NULL
    AND bmi IS NOT NULL
    AND ckd IS NOT NULL
)
SELECT
  RANK() OVER (ORDER BY 1000 * SUM(death) / (SUM(time_months) / 12) DESC) AS risk_rank,
  COALESCE(NULLIF(profile, ''), 'None of the four') AS profile,
  COUNT(*) AS participants,
  SUM(death)::INT AS deaths,
  ROUND(1000 * SUM(death) / (SUM(time_months) / 12), 2) AS deaths_per_1000_py
FROM profiles
GROUP BY profile
HAVING COUNT(*) >= 200
ORDER BY risk_rank
```

Table 3.3 Top six and bottom three of 16 profiles (q04).

Rank	Profile	Participants	Deaths	Deaths per 1,000 PY
1	Diabetes + Hypertension + CKD	1,092	573	78.71
2	Hypertension + CKD	2,067	1,057	70.82
3	Diabetes + CKD	320	126	52.31
4	Diabetes + Hypertension + Obesity + CKD	1,469	554	51.35
5	Hypertension + Obesity + CKD	1,173	407	42.33
6	Diabetes + Obesity + CKD	330	96	38.82
14	Diabetes + Obesity	958	115	14.01
15	None of the four	17,241	1,090	6.04
16	Obesity	6,939	315	4.61

Diabetes with hypertension and CKD has the highest crude rate at 78.71 per 1,000 person-years, 13 times the rate of people with none of the four. Every one of the top six profiles includes CKD. The lowest is obesity alone, even below "none of the four". That looks odd, and Section 4.5 shows why: people with obesity alone are younger.

3.3 Care gaps

Table 3.4 Hypertension care cascade by cycle (q08).

Cycle	With hypertension	Aware %	Treated %	Controlled %
1999-2000	1,577	69.4	60.0	26.7
2001-2002	1,681	72.0	62.6	32.0
2003-2004	1,753	77.0	68.2	37.2
2005-2006	1,571	78.9	70.3	42.1
2007-2008	2,173	80.3	73.6	46.3
2009-2010	2,180	83.5	77.3	50.3
2011-2012	1,892	83.3	76.1	49.3
2013-2014	2,008	85.1	76.6	51.8
2015-2016	2,002	82.7	74.9	46.2
2017-2018	2,167	79.4	72.1	41.5

Table 3.5 Undiagnosed and poorly controlled diabetes (q09).

Cycle	With diabetes	Undiagnosed %	HbA1c 9+ % (diagnosed)
1999-2000	577	23.7	22.3
2001-2002	626	26.8	16.8
2003-2004	649	21.9	14.6
2005-2006	634	24.1	16.6
2007-2008	1,016	28.9	14.6
2009-2010	1,001	28.8	14.5
2011-2012	929	27.3	17.5
2013-2014	945	25.4	16.2
2015-2016	1,075	25.1	17.6
2017-2018	1,123	25.9	15.5

Blood pressure control among people with hypertension rose from 26.7% in 1999-2000 to 51.8% in 2013-2014, then fell back to 41.5% by 2017-2018. Awareness and treatment followed the same rise and fall. For diabetes, about a quarter of cases were undiagnosed in every cycle (21.9% to 28.9%), with no clear improvement over 20 years.

3.4 Kidney disease grid and life table

Table 3.6 Crude deaths per 1,000 person-years by KDIGO eGFR stage (rows) and albuminuria stage (columns). Cells under 30 people are blank (q10).

eGFR stage	A1	A2	A3
G1	5.9	16.7	27.4
G2	20.4	54.6	55.4
G3a	51.9	93.5	103.4
G3b	73.6	110.0	115.7
G4	109.0	122.4	153.4
G5	nan	nan	107.3

The KDIGO grid nephrologists use shows the expected pattern: risk rises both down the eGFR stages and across the albuminuria stages. Albuminuria matters even when eGFR is normal. In stage G1, people with A2 or A3 albuminuria die at about 3 and 5 times the rate of people with A1.

Code 3.3 An actuarial life table built with window functions (q11). Censored people count as at risk for half the year. 03_duckdb_sql.ipynb, cell 23

```
WITH yearly AS (
  SELECT
    glycemc_status,
    FLOOR(time_months / 12)::INT AS fu_year,
    COUNT(*) AS n_exit,
    SUM(death) AS n_death
  FROM cohort
  WHERE glycemc_status IS NOT NULL
  GROUP BY glycemc_status, fu_year
),
at_risk AS (
  SELECT
    glycemc_status,
    fu_year,
    n_death,
    n_exit - n_death AS n_censored,
    SUM(n_exit) OVER (PARTITION BY glycemc_status ORDER BY fu_year DESC) AS n_at_risk
  FROM yearly
)
SELECT
  glycemc_status,
  fu_year,
  n_at_risk::INT AS n_at_risk,
  n_death::INT AS n_death,
  ROUND(100 * EXP(SUM(LN(GREATEST(1 - n_death / (n_at_risk - n_censored / 2), 1e-9))))
        OVER (PARTITION BY glycemc_status ORDER BY fu_year)), 1) AS pct_surviving
  FROM at_risk
ORDER BY glycemc_status, fu_year
```

Table 3.7 Crude percent surviving by glycemc status, from the SQL life table.

Years	normal	prediabetes	diabetes
5	95.9	93.1	85.9
10	90.5	83.5	69.4
15	84.6	72.6	53.6
20	77.8	60.7	38.0

After 20 years, 78% of people with normal glucose were alive, against 61% with prediabetes and 38% with diabetes. The weighted Kaplan-Meier curves in Section 6 replace this crude version.

4 Exploratory analysis

Notebook 04. Checks that shape the R models: skewed measures, overlapping biomarkers, missing data, risk shapes, the 2007 jump in prediabetes, and how much age distorts crude comparisons.

The EDA answers six questions, each tied to a decision in the R scripts. Everything here is still crude and unweighted.

Table 4.1 EDA questions and the decisions they feed.

Question	Finding	Decision for the R models
Which measures are skewed?	ACR (skew 95), triglycerides, HbA1c, fasting glucose, BMI, SBP, HDL	ACR modelled on the log scale; biomarkers as splines
Did the sample mix change between cycles?	Hispanic share jumps in 2007, Asian share from 2011; HbA1c also steps up within every group	Survey weights, cycle as a covariate, sensitivity analysis without it
Which biomarkers overlap?	Total with non-HDL cholesterol 0.91, BMI with waist 0.89, age with eGFR -0.71	Keep BMI (less missing), total cholesterol only in model C
Is missing data random?	People missing BMI, ACR or waist die at about 3 to 4 times the rate	Multiple imputation with the outcome in the model
Is risk linear?	U or J shapes for HbA1c, BMI, SBP, eGFR, HDL	Restricted cubic splines
How much does age distort crude rates?	Prediabetes excess disappears; current smoking risk doubles	Age adjustment in every comparison

4.1 Distributions

Table 4.2 Measures with skewness above 1, before and after a log transform.

variable	n	median	p1	p99	skewness	skewness_log
acr	49,456	7.12	1.61	877.06	95.27	1.81
tg	22,945	107.00	31.44	528.00	9.57	0.43
fpg	23,357	100.00	75.00	291.00	4.32	2.35
hba1c	48,283	5.50	4.50	10.90	3.62	2.38
bmi	49,782	27.87	17.90	50.70	1.32	0.41
hdl	47,697	50.00	26.00	101.00	1.12	0.06
sbp	48,485	122.00	92.67	187.33	1.10	0.56

Urine ACR is extremely skewed (skewness 95), and its log is close to symmetric, so it enters every model as $\log(\text{ACR})$. HbA1c and fasting glucose stay skewed even after a log transform, because of the long diabetes tail. Splines handle that better than a transform.

4.2 Did the sample change between cycles?

In the SQL trend table, crude prediabetes jumped from 27.6% in 2005-2006 to 35.2% in 2007-2008. There are two possible reasons. One is composition: from 2007 NHANES oversampled all Hispanic adults, not only Mexican Americans, and from 2011 Asian adults. The other is measurement, a change in the HbA1c lab method.

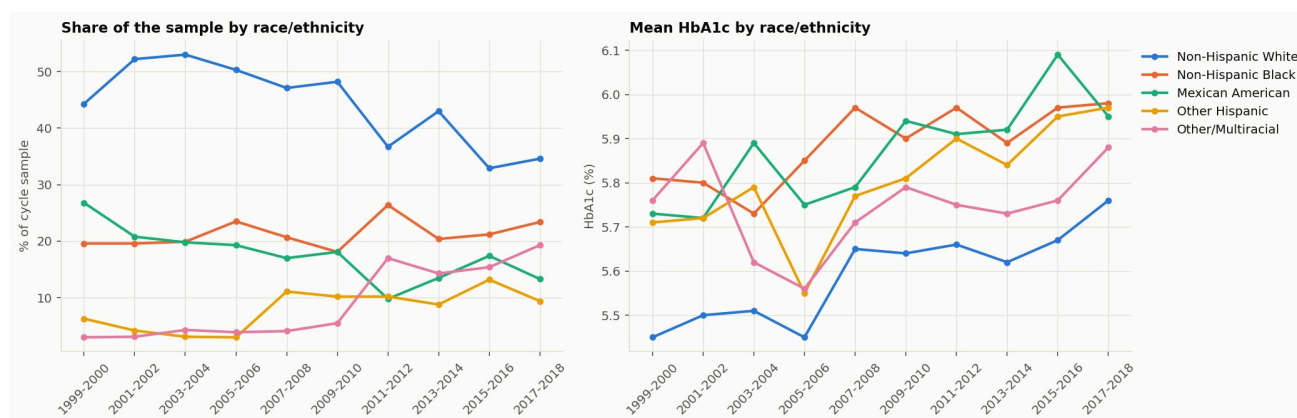


Figure 4.1 Race and ethnicity mix of the sample (left) and mean HbA1c within each group (right), by cycle.

Both happened. The Other Hispanic share rose from 3.0% to 11.1% in 2007-2008, and Other/Multiracial from 5.5% to 17.0% in 2011-2012. But mean HbA1c also stepped up within groups in 2007-2008, for example from 5.45 to 5.65 in Non-Hispanic White adults and from 5.55 to 5.77 in Other Hispanic adults. Composition cannot explain a jump within a group, so part of it is measurement. Survey weights fix the composition part. For the measurement part, every R model includes survey cycle as a covariate, and a sensitivity analysis drops it (Section 6.6).

4.3 Overlapping biomarkers

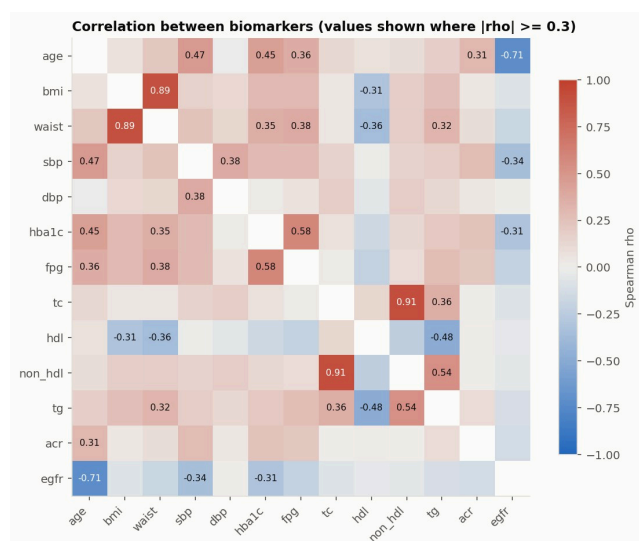


Figure 4.2 Spearman correlation between biomarkers.

Three pairs have $|\rho|$ of 0.7 or more. Total and non-HDL cholesterol (0.91) are close to the same number, since non-HDL is total minus HDL. BMI and waist (0.89) measure the same thing, and BMI is kept because it is missing less often (2.0% vs 5.8%). Age and eGFR (-0.71) are linked by construction, since age is in the eGFR equation.

Putting both members of a pair in one model makes their coefficients unstable, so each model uses only one of them.

4.4 Is missing data random?

Table 4.3 People missing a value compared with people who have it. Death rates per 1,000 person-years.

variable	n missing	% missing	age, missing	age, observed	rate, missing	rate, observed	rate ratio
bmi	1,037	2.0	60.6	50.1	62.2	16.5	3.77
education	81	0.2	64.6	50.3	59.5	17.2	3.46
acr	1,363	2.7	57.0	50.1	53.2	16.4	3.24
waist	2,935	5.8	57.3	49.8	46.4	15.9	2.92
smoking	51	0.1	57.8	50.3	27.9	17.2	1.62
hdl	3,122	6.1	50.4	50.3	25.5	16.7	1.53
tc	3,120	6.1	50.4	50.3	25.4	16.7	1.52
egfr	3,248	6.4	50.5	50.3	25.1	16.7	1.50
hba1c	2,536	5.0	49.0	50.3	24.6	16.9	1.46
sbp	2,334	4.6	49.8	50.3	22.8	17.0	1.34
pir	4,592	9.0	52.6	50.1	19.7	17.0	1.16

The answer is no. People with no BMI are 10 years older and die at 3.8 times the rate of people with a BMI. The same holds for ACR (3.2 times) and waist (2.9 times). Many of these are frail people who could not be weighed or measured. Dropping them would leave a healthier sample and bias every hazard ratio. This is the main reason the R work uses multiple imputation instead of complete-case analysis.

4.5 How much does age distort crude rates?

Each group's age-specific death rates are re-weighted to the age mix of the whole cohort. The result answers the question "what would this group's death rate be if it had the same ages as everyone else?"

Code 4.1 Crude and age-standardized death rates in one function.

04_eda.ipynb, cell 3

```
def rates(df, by):
    a = df.groupby(by, observed=True).agg(n=("SEQN", "size"), deaths=("death", "sum"), py=("time_months", lambda s:
s.sum() / 12))
    return a.assign(rate=1000 * a.deaths / a.py)

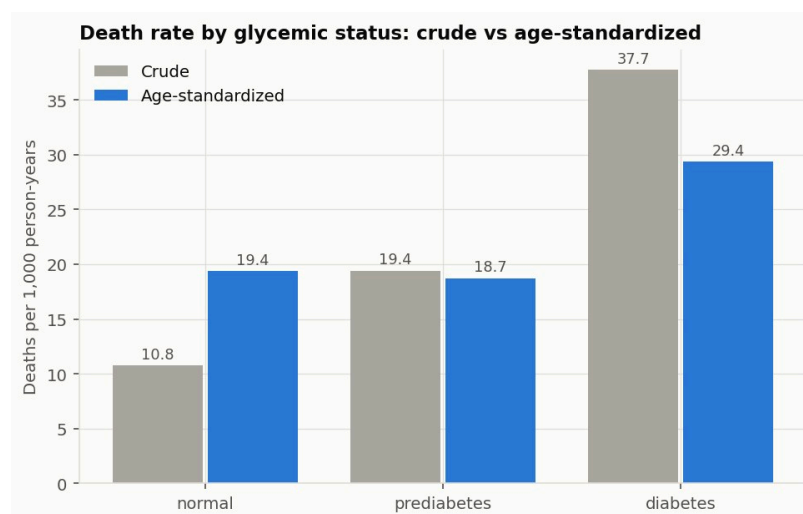
def crude_vs_std(df, by):
    strat = rates(df, [by, "age_group"]).rate.unstack()
    w = df.age_group.value_counts(normalize=True).reindex(strat.columns)
    return rates(df, by).assign(age_std_rate=(strat * w).sum(axis=1, min_count=1) /
strat.notna().mul(w).sum(axis=1)).round(2)
```

Table 4.4 Crude vs age-standardized death rate per 1,000 person-years.

Grouping	Level	n	deaths	Crude	Age-standardized
Glycemic	diabetes	8,575	2,520	37.7	29.4
Glycemic	normal	24,885	2,808	10.8	19.4
Glycemic	prediabetes	15,175	2,555	19.4	18.7
Conditions	0	18,540	1,380	7.2	16.6
Conditions	1	16,053	2,927	19.7	19.6
Conditions	2	8,544	2,051	28.0	23.3
Conditions	3	3,280	880	34.3	28.4
BMI	underweight	816	183	25.0	39.1
BMI	normal	14,042	2,271	16.7	21.9
BMI	overweight	16,807	2,736	16.9	19.0
BMI	obese	18,117	2,574	15.5	20.2
Smoking	current	10,781	1,692	15.9	31.9
Smoking	former	12,592	3,214	28.2	23.1
Smoking	never	27,395	3,358	13.0	16.9
Sex	female	25,667	3,687	15.1	17.9
Sex	male	25,152	4,592	19.5	25.0

Age changes the picture in three places.

- **Prediabetes.** Crude, it looks almost twice as deadly as normal glucose (19.4 vs 10.8). Age-standardized, it is about the same (18.7 vs 19.4). The crude gap was age.
- **Smoking.** The crude rate hides the risk, because current smokers are younger. Age-standardized, current smokers die at 31.9 per 1,000 person-years against 16.9 for never smokers.
- **BMI.** The shape is U or J: underweight is highest (39.1), overweight is lowest (19.0), and obesity is close to normal weight.

**Figure 4.3** Death rate by glycemic status, crude and age-standardized.

4.6 Is risk a straight line or a curve?

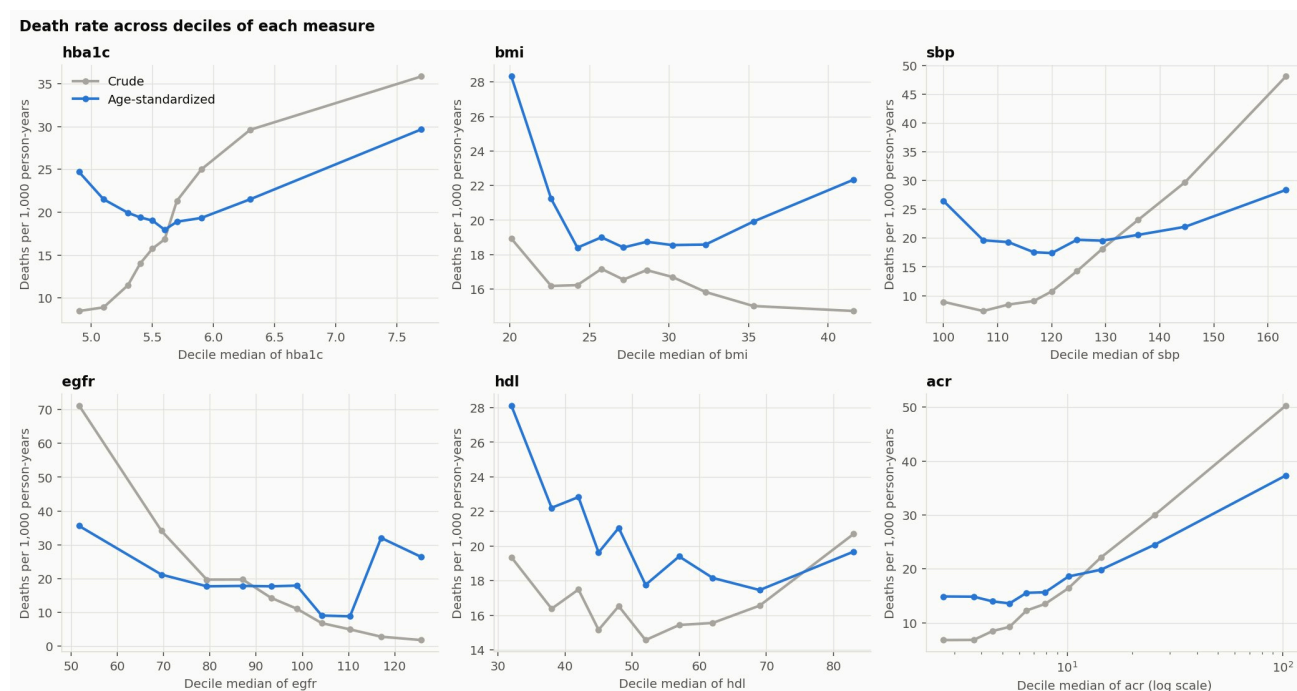


Figure 4.4 Crude (grey) and age-standardized (blue) death rate across deciles of six measures.

After age standardization, HbA1c, BMI, systolic blood pressure, eGFR and HDL all show U or J shapes: risk is higher at both ends. A single linear term would miss that, so these measures enter the R models as restricted cubic splines. ACR is closer to a straight line on the log scale. The top eGFR deciles show an uptick in risk, which is partly unstable because the strata are sparse. Section 6.4 checks it with continuous age adjustment.

5 Survey design, imputation and prevalence

Script 05. Setting up the survey design, filling missing values with multiple imputation, and estimating how common each condition is in US adults.

5.1 Survey design

The design uses the masked variance units NHANES provides: 148 strata and 301 primary sampling units (PSUs), with the combined 10-cycle exam weight. Strata numbers are unique across cycles, so the cycles stack with `nest = TRUE`. A subtle point: analysing a subset of a survey as if it were the whole survey can give wrong standard errors. Here the cohort keeps all 301 PSUs of the full sample, with at least 60 people in each, so a design built on the cohort gives the same standard errors as a full-sample domain analysis. The design has 153 degrees of freedom.

5.2 Multiple imputation

11,212 people (22.1%) have at least one missing value among the 15 variables used in the models. They are older (mean 53.0 vs 49.5) and die at almost twice the rate (27.1 vs 14.7 per 1,000 person-years). Complete-case analysis would throw them away.

Missing values are filled with multiple imputation by chained equations (`mice`): 20 imputed datasets, 10 iterations each, run in parallel with seed 2026. Continuous and binary variables use predictive mean matching, so imputed values are always real observed values. Education and smoking use multinomial regression. The imputation model includes the death indicator and the Nelson-Aalen cumulative hazard, as White and Royston (2009) recommend for survival analysis, so imputed values carry the link with the outcome instead of weakening it. It also includes age, sex, race, cycle and the log of the survey weight. Fasting-only measures and skip-pattern medication questions are not imputed.

Code 5.1 Survey design, then imputation with the outcome and cumulative hazard in the model.

`r/05_survey_design_imputation.R`, lines 96-96, lines 132-143

```
des_obs <- svydesign(ids = ~psu, strata = ~strata, weights = ~wt_mec, nest = TRUE, data = cohort)
...
imp_data <- cohort %>%
  mutate(na_hazard = nelsonaalen(cohort, time_months, death), log_wt = log(wt_mec)) %>%
  select(all_of(imp_vars), age, sex, race_eth, cycle, death, na_hazard, log_wt)

imp_file <- file.path(out, "05_mice.rds")
if (file.exists(imp_file)) {
  cat("Loading cached imputation...\n")
  imp <- readRDS(imp_file)
} else {
  cat("Running imputation in parallel (this takes a while)...\n")
  imp <- futuremice(imp_data, m = 20, maxit = 10, parallelseed = 2026)
  saveRDS(imp, imp_file)
```

After imputation, the conditions are rebuilt from the completed values with the same rules as notebook 02. Observed statuses are kept and only the missing ones are filled. All later estimates are computed in each of the 20 datasets and pooled with Rubin's rules, so the confidence intervals include the extra uncertainty from imputation.

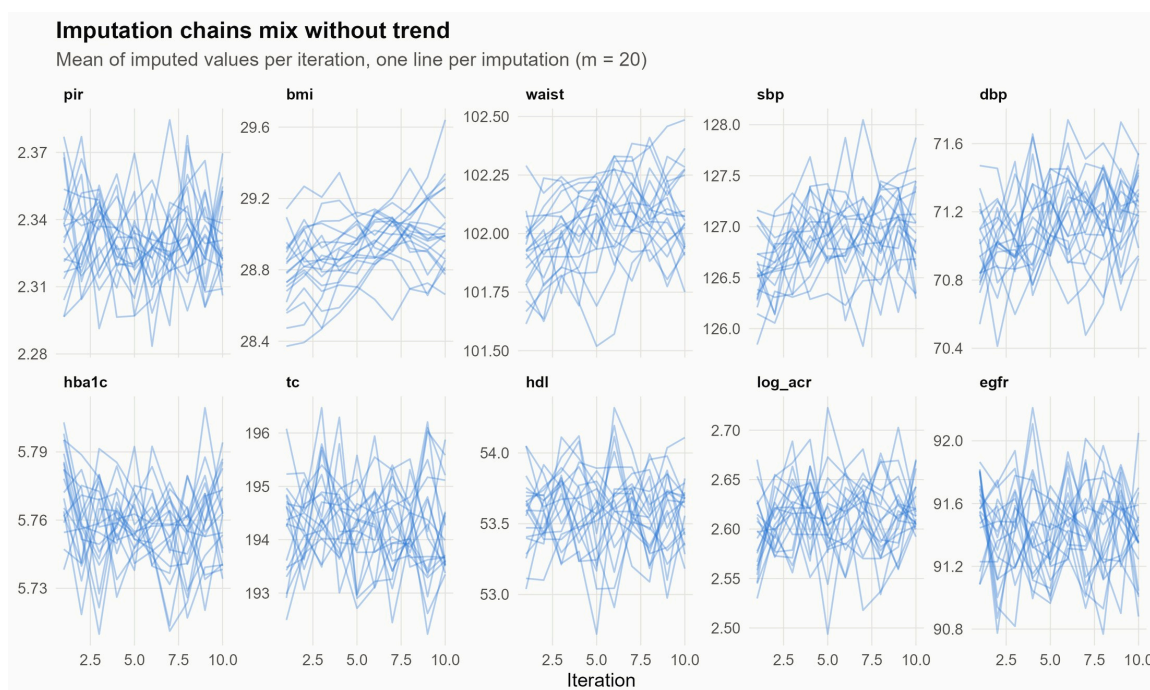


Figure 5.1 Mean of the imputed values per iteration, one line per imputation. The chains mix without a trend.

The imputed values are close to the observed ones for most variables. Where they differ, they differ in the expected direction: imputed eGFR is lower (91.39 vs 93.39) and imputed waist is larger (102.08 vs 98.68 cm), because the people missing them are older and sicker. That is the point of including the outcome in the imputation model.

5.3 Prevalence in US adults

Prevalence is the survey-weighted share of adults with each condition. The age-standardized version uses the direct method with the 2000 US standard population in three age groups (20-39, 40-59, 60+), the same standard CDC uses.

Code 5.2 Crude and age-standardized prevalence, pooled over imputations.

r/05_survey_design_imputation.R, lines 202-202, lines 210-212

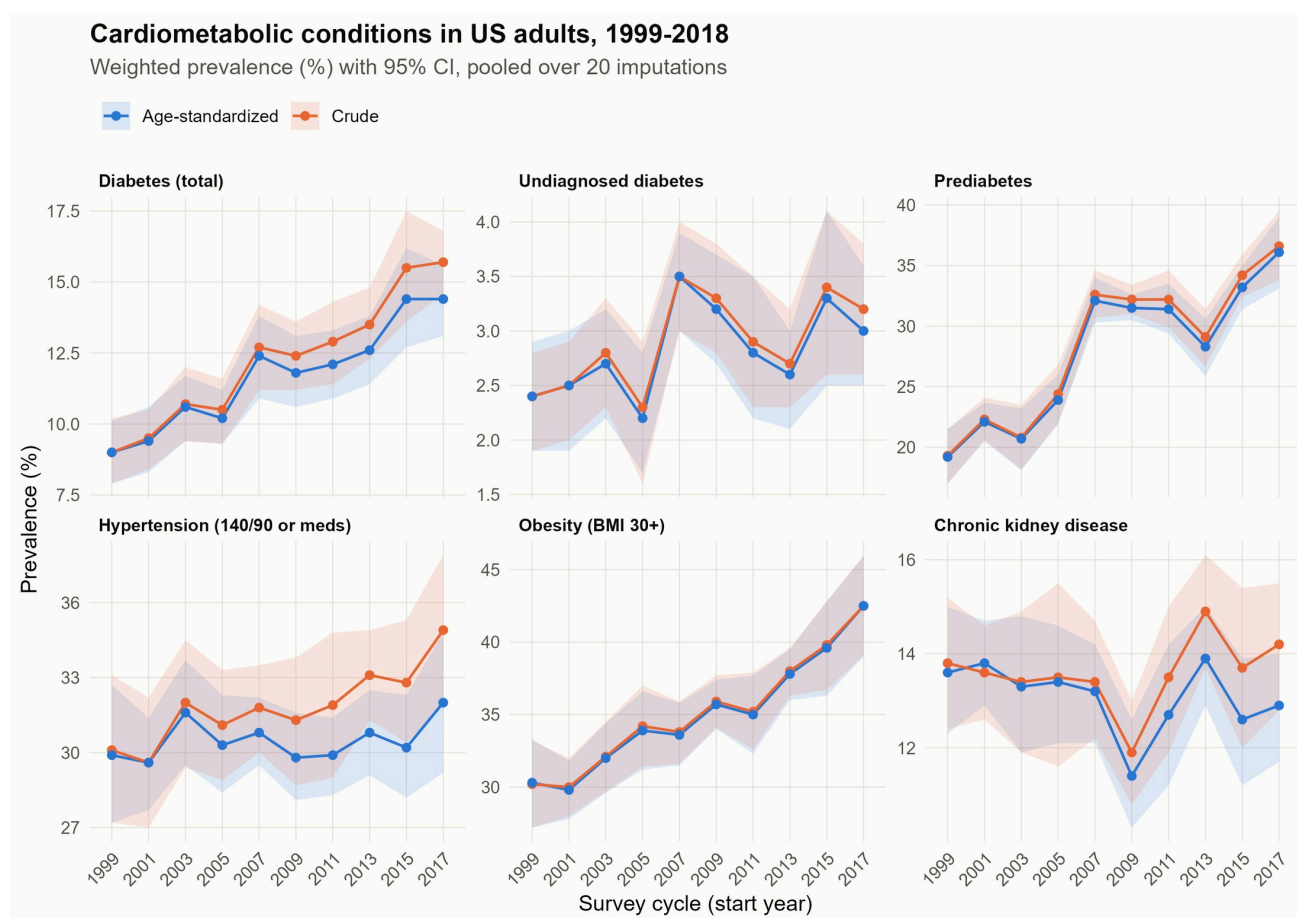
```
des <- svydesign(ids = ~psu, strata = ~strata, weights = ~wt_mec, nest = TRUE, data = imputationList(split(long,
long$.imp)))
...
prev_overall <- bind_rows(
  with(des, fun = function(d) svymean(f_conds, d)) %>% mi_tidy() %>% mutate(condition = term, estimate = "Crude (MI)"),
  with(des, fun = function(d) svymean(f_conds, svystandardize(d, ~age3, ~1, std_pop$pop))) %>% mi_tidy() %>%
  mutate(condition = term, estimate = "Age-standardized (MI)"),
```

Table 5.1 Weighted prevalence in US adults aged 20+, 1999-2018, pooled over 20 imputations, and the trend across cycles.

Condition	Age-standardized % (95% CI)	Crude % (95% CI)	OR per cycle (trend)	p trend
Diabetes (total)	11.8 (11.4 to 12.2)	12.4 (12.0 to 12.8)	1.056	<0.001
Diagnosed diabetes	9.0 (8.7 to 9.4)	9.5 (9.1 to 9.8)	1.064	<0.001
Undiagnosed diabetes	2.8 (2.6 to 3.0)	2.9 (2.7 to 3.1)	1.017	0.153
Prediabetes	28.2 (27.5 to 28.8)	28.7 (28.0 to 29.4)	1.089	<0.001
Hypertension (140/90 or meds)	30.5 (29.8 to 31.1)	31.9 (31.1 to 32.7)	1.005	0.575
Hypertension (130/80 or meds)	45.7 (45.0 to 46.5)	47.2 (46.4 to 48.0)	0.984	0.030
Obesity (BMI 30+)	35.2 (34.4 to 36.1)	35.4 (34.6 to 36.2)	1.062	<0.001
Chronic kidney disease	13.0 (12.6 to 13.4)	13.6 (13.1 to 14.1)	0.987	0.050
Albuminuria (ACR 30+)	9.7 (9.3 to 10.1)	10.0 (9.6 to 10.4)	0.987	0.076
Current smoking	22.1 (21.3 to 22.8)	21.7 (21.0 to 22.4)	0.954	<0.001
Self-reported CVD	8.3 (8.0 to 8.7)	8.9 (8.5 to 9.3)	0.981	0.031

Trend: survey logistic regression of each condition on cycle number (1 to 10), adjusted for age group, sex and race/ethnicity. The OR is the change in odds per 2-year cycle.

Over 1999-2018, the age-standardized prevalence was 11.8% for diabetes, 28.2% for prediabetes, 30.5% for hypertension, 35.2% for obesity and 13.0% for CKD. About 23.4% of all diabetes was undiagnosed (a crude prevalence of 2.9% out of 12.4%). Under the newer ACC/AHA definition of 130/80, hypertension goes up to 45.7%.

**Figure 5.2** Weighted prevalence by survey cycle, age-standardized and crude, with 95% CI.

Diabetes, prediabetes and obesity all rose over the 20 years. Age-standardized diabetes went from 9.0% in 1999-2000 to 14.4% in 2017-2018, and obesity from 30.3% to 42.5%. The rise in diabetes came from diagnosed cases (OR 1.064 per cycle), while undiagnosed diabetes showed no clear trend ($p = 0.153$). Hypertension at 140/90 was flat. Current smoking fell

from 24.4% to 18.0%. The prediabetes jump in 2007-2008 is still there after weighting (23.9% to 32.1%), which fits the measurement change found in Section 4.2. Its trend estimate should be read with that in mind.

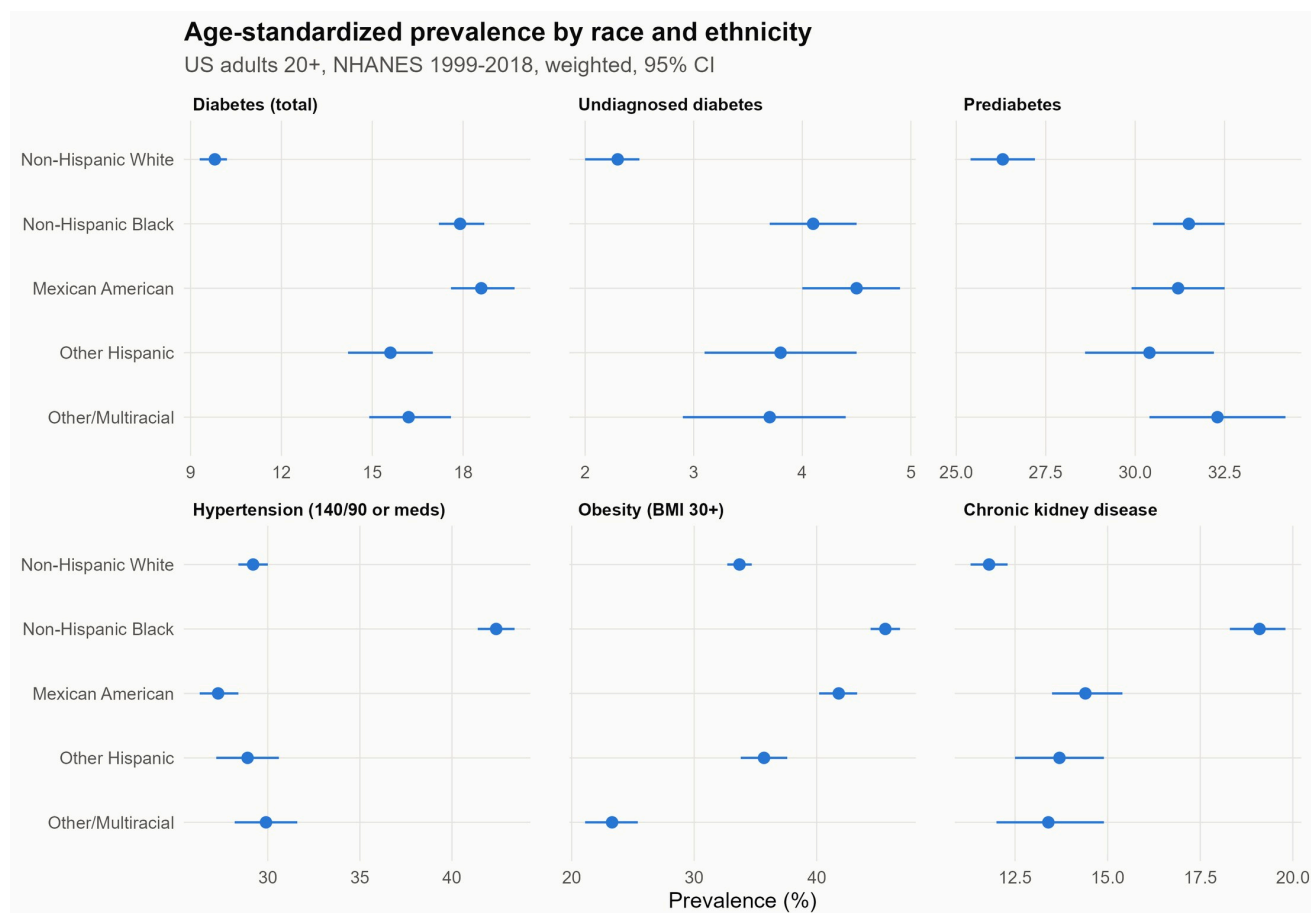


Figure 5.3 Age-standardized prevalence by race and ethnicity, with 95% CI.

The differences between groups are large. Age-standardized diabetes was 9.8% in Non-Hispanic White adults, 17.9% in Non-Hispanic Black adults and 18.6% in Mexican American adults. Non-Hispanic Black adults also had the highest hypertension (42.4%), obesity (45.6%) and CKD (19.1%). Education shows a steady gradient: diabetes was 16.2% in adults without a high school diploma against 10.2% in those with more than high school.

5.4 Baseline characteristics

Table 5.2 Weighted baseline characteristics by glycemic status (Table 1). Means with 95% CI, or weighted %.

Characteristic	Normal	Prediabetes	Diabetes	Overall
Participants, n (unweighted)	26443	15718	8658	50819
Deaths, n (unweighted)	3059	2676	2544	8279
Age, years	42.2 (41.9-42.6)	52.5 (52.1-53.0)	59.2 (58.8-59.7)	47.3 (47.0-47.7)
Sex: female	53.8%	47.1%	48.6%	51.2%
Race/ethnicity: Non-Hispanic Black	9.8%	12.2%	15.5%	11.2%
Race/ethnicity: Mexican American	7.7%	8.4%	9.2%	8.1%
Education: <HS	14.7%	19.8%	26.0%	17.6%
Smoking: current	22.8%	21.4%	16.9%	21.7%
BMI, kg/m ²	27.3 (27.2-27.4)	30.0 (29.8-30.1)	32.9 (32.6-33.1)	28.8 (28.6-28.9)
Systolic BP, mmHg	119.5 (119.2-119.9)	126.4 (125.9-126.8)	131.2 (130.6-131.9)	122.9 (122.6-123.3)
HbA1c, %	5.2 (5.2-5.2)	5.7 (5.6-5.7)	7.2 (7.2-7.3)	5.6 (5.6-5.6)
HDL cholesterol, mg/dL	54.9 (54.6-55.3)	51.7 (51.3-52.1)	47.4 (47.0-47.9)	53.1 (52.8-53.4)
eGFR, mL/min/1.73m ²	99.5 (99.0-100.0)	92.0 (91.5-92.5)	85.0 (84.3-85.7)	95.5 (95.1-96.0)
Urine ACR, mg/g (geometric mean)	7.0 (6.9-7.1)	8.3 (8.1-8.5)	18.4 (17.7-19.2)	8.3 (8.2-8.4)
Hypertension	20.7%	40.5%	65.5%	31.9%
Chronic kidney disease	8.5%	14.5%	35.6%	13.6%
Self-reported CVD	4.8%	10.3%	24.9%	8.9%
Self-reported cancer	7.4%	11.8%	15.7%	9.7%

Counts are unweighted and averaged over imputations. Selected rows; the full table is in `outputs/R/05_survey_design_imputation/05_table1_weighted.csv`.

Adults with diabetes were on average 17 years older than adults with normal glucose (59.2 (58.8-59.7) vs 42.2 (41.9-42.6)), had a higher BMI and blood pressure, lower eGFR, and more than twice the ACR. 65.5% had hypertension and 35.6% had CKD. That overlap is why the Cox models in Section 6 adjust for the conditions together.

6 Survival and hazard ratios

Script 06. Weighted death rates, Kaplan-Meier survival, Cox models with three levels of adjustment, dose-response splines, a check of proportional hazards, and six sensitivity analyses.

6.1 Weighted death rates

The crude weighted rate is the survey ratio of deaths to person-years. The age-standardized rate combines age-specific rates for 20-39, 40-59 and 60+ with the 2000 standard population weights (0.3966, 0.3718, 0.2316), and its standard error comes from a survey contrast of the age-specific rates. Diabetes is split into diagnosed and undiagnosed from here on.

Table 6.1 Survey-weighted deaths per 1,000 person-years, pooled over 20 imputations.

Group	Level	Crude (95% CI)	Age-standardized (95% CI)
Glycemic status	normal	7.6 (7.1 to 8.0)	11.6 (10.9 to 12.2)
	prediabetes	15.1 (14.2 to 16.0)	12.1 (11.4 to 12.9)
	undiagnosed diabetes	27.1 (24.2 to 30.1)	18.2 (16.2 to 20.2)
	diagnosed diabetes	34.6 (32.6 to 36.7)	23.0 (21.0 to 25.1)
Hypertension	no hypertension	6.0 (5.7 to 6.3)	10.0 (9.4 to 10.5)
	hypertension	27.0 (25.9 to 28.1)	16.8 (16.0 to 17.6)
Chronic kidney disease	no CKD	8.2 (7.8 to 8.5)	9.8 (9.4 to 10.3)
	CKD	42.6 (40.7 to 44.6)	25.9 (24.6 to 27.1)
BMI category	normal	11.2 (10.5 to 11.9)	14.5 (13.6 to 15.3)
	underweight	17.5 (14.0 to 21.0)	27.6 (22.2 to 32.9)
	overweight	11.9 (11.2 to 12.6)	11.9 (11.2 to 12.5)
	obese	12.5 (11.9 to 13.2)	12.6 (12.0 to 13.3)
Smoking	never	8.9 (8.4 to 9.4)	10.4 (9.8 to 10.9)
	former	18.8 (17.6 to 20.0)	13.6 (12.8 to 14.4)
	current	12.2 (11.4 to 12.9)	18.6 (17.6 to 19.7)

After age standardization, the death rate was 11.6 per 1,000 person-years with normal glucose, 12.1 with prediabetes, 18.2 with undiagnosed diabetes and 23.0 with diagnosed diabetes. The prediabetes interval overlaps normal glucose. CKD more than doubled the age-standardized rate (25.9 vs 9.8). Underweight adults had the highest rate of any BMI group (27.6), and overweight the lowest (11.9). For smoking the crude order is misleading: former smokers look worst crude (18.8) because they are older, while current smokers are worst after age standardization (18.6).

6.2 Kaplan-Meier survival

Weighted Kaplan-Meier curves are estimated in each imputation with `svykm` and averaged on a common time grid. They stop at 18 years, because only the 1999-2000 cycle is followed longer and the tail rests on a few people.

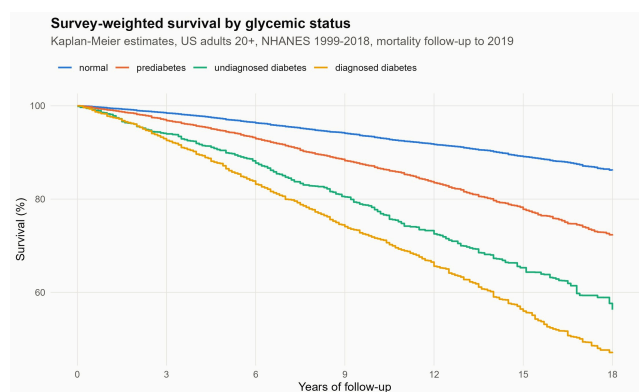


Figure 6.1 Weighted survival by glycemic status.

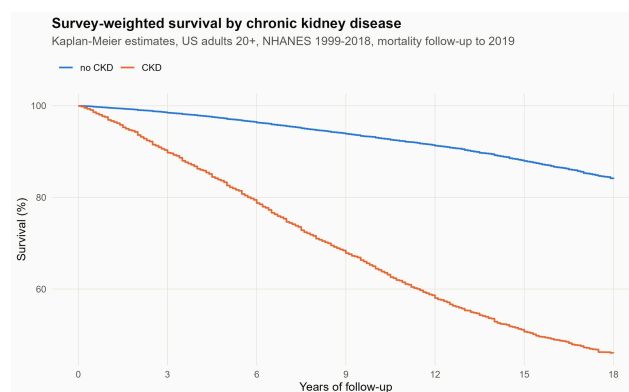


Figure 6.2 Weighted survival by CKD status.

Table 6.2 Weighted percent surviving at 5, 10 and 15 years.

Group	Level	5 years	10 years	15 years
Glycemic status	normal	97.0	93.2	89.1
	prediabetes	94.4	86.8	77.8
	undiagnosed diabetes	89.9	77.7	65.3
	diagnosed diabetes	86.5	71.6	56.0
Hypertension	no hypertension	97.6	94.7	91.2
	hypertension	89.8	77.2	64.7
Chronic kidney disease	no CKD	97.1	93.0	88.0
	CKD	82.6	64.4	50.7
BMI category	normal	95.0	89.7	84.3
	underweight	91.6	83.0	76.1
	overweight	95.5	89.3	82.8
	obese	95.1	89.0	82.2

At 15 years, 89.1% of adults with normal glucose were alive, against 56.0% with diagnosed diabetes and 50.7% with CKD. These curves are not age-adjusted, so they mix the effect of the condition with the effect of being older. The Cox models separate the two.

6.3 Cox models

Survey-weighted Cox proportional hazards models (`svycoxph`) are fitted in each of the 20 imputed datasets and pooled with Rubin's rules. Time is years since the exam. Age enters as a natural spline with knots at 35, 50 and 65, because the log hazard is not linear in age. Adjustment is added in three steps, so the change between steps shows how much of each association runs through the added variables.

- **M1:** age and sex.
- **M2:** M1 plus race and ethnicity, education, income-to-poverty ratio, smoking and survey cycle.
- **M3:** M2 plus all conditions together: glycemic status, hypertension, CKD, BMI group, prior CVD and cancer.

Code 6.1 The three adjustment sets and the pooled survey Cox fit.

r/06_survival_cox.R, lines 160-165

```
age_term <- "ns(age, knots = c(35, 50, 65), Boundary.knots = c(20, 85))"
m1 <- c(age_term, "sex")
m2 <- c(m1, "race_eth", "education", "pir", "smoking", "cycle")
m3 <- c(m2, "glycemic", "hypertension", "ckd", "bmi_cat", "cvd_history", "cancer")
cox_f <- function(terms, time = "time_y", event = "death") reformulate(unique(terms), sprintf("Surv(%s, %s)", time,
event))
fit_mi <- function(f, design = des) with(design, fun = function(d) svycoxph(f, design = d)) %>% mi_tidy()
```

Table 6.3 All-cause mortality hazard ratios (95% CI) by adjustment level.

Exposure	M1: age, sex	M2: + socio, smoking, cycle	M3: + all conditions
Prediabetes	1.06 (0.98 to 1.13)	1.02 (0.95 to 1.10)	0.99 (0.92 to 1.06)
Diabetes	1.81 (1.70 to 1.93)	1.70 (1.59 to 1.83)	1.44 (1.34 to 1.55)
Undiagnosed diabetes	1.52 (1.35 to 1.70)	1.43 (1.26 to 1.61)	not fitted
Diagnosed diabetes	1.91 (1.78 to 2.05)	1.80 (1.67 to 1.94)	not fitted
Hypertension	1.41 (1.32 to 1.51)	1.38 (1.29 to 1.47)	1.22 (1.14 to 1.31)
CKD	1.96 (1.85 to 2.07)	1.83 (1.73 to 1.93)	1.59 (1.50 to 1.68)
Underweight	2.10 (1.70 to 2.61)	1.61 (1.31 to 1.99)	1.57 (1.20 to 2.06)
Overweight	0.84 (0.78 to 0.91)	0.88 (0.82 to 0.94)	0.83 (0.78 to 0.89)
Obese	1.10 (1.03 to 1.17)	1.12 (1.05 to 1.20)	0.93 (0.87 to 1.00)

Reference groups: normal glucose, no hypertension, no CKD, normal BMI. The diagnosed and undiagnosed split replaces the diabetes term in its own M1 and M2 models; in M3 all conditions enter together with diabetes as one group.

In the fully adjusted model M3, diabetes was linked to a 44% higher rate of death (HR 1.44, 95% CI 1.34 to 1.55), CKD to 59% (HR 1.59, 95% CI 1.50 to 1.68) and hypertension to 22% (HR 1.22, 95% CI 1.14 to 1.31). Prediabetes had no excess risk at any level of adjustment (M3 HR 0.99, 95% CI 0.92 to 1.06). This confirms what the age-standardized rates suggested in Section 4.5.

The diabetes HR drops from 1.81 in M1 to 1.44 in M3. The drop from M2 to M3 comes from adding the other conditions, including hypertension and CKD. These are partly consequences of diabetes, so M3 shows the association of diabetes over and above the damage it has already done to blood pressure and kidneys. Section 8 estimates the total effect separately. People with diagnosed diabetes had a higher risk than people with undiagnosed diabetes (1.80 vs 1.43 in M2), most likely because diagnosed diabetes has usually been present for longer and is more advanced.

Obesity was not linked to higher risk once the conditions were in the model (M3 HR 0.93, 95% CI 0.87 to 1.00), and overweight was linked to lower risk (HR 0.83, 95% CI 0.78 to 0.89). This is a common finding in cohort data and does not mean that weight gain is protective. Obesity acts largely through diabetes, hypertension and CKD, which M3 adjusts for, and people who lose weight because of illness can make the normal-weight group look worse. Underweight had a clear excess risk (HR 1.57, 95% CI 1.20 to 2.06).

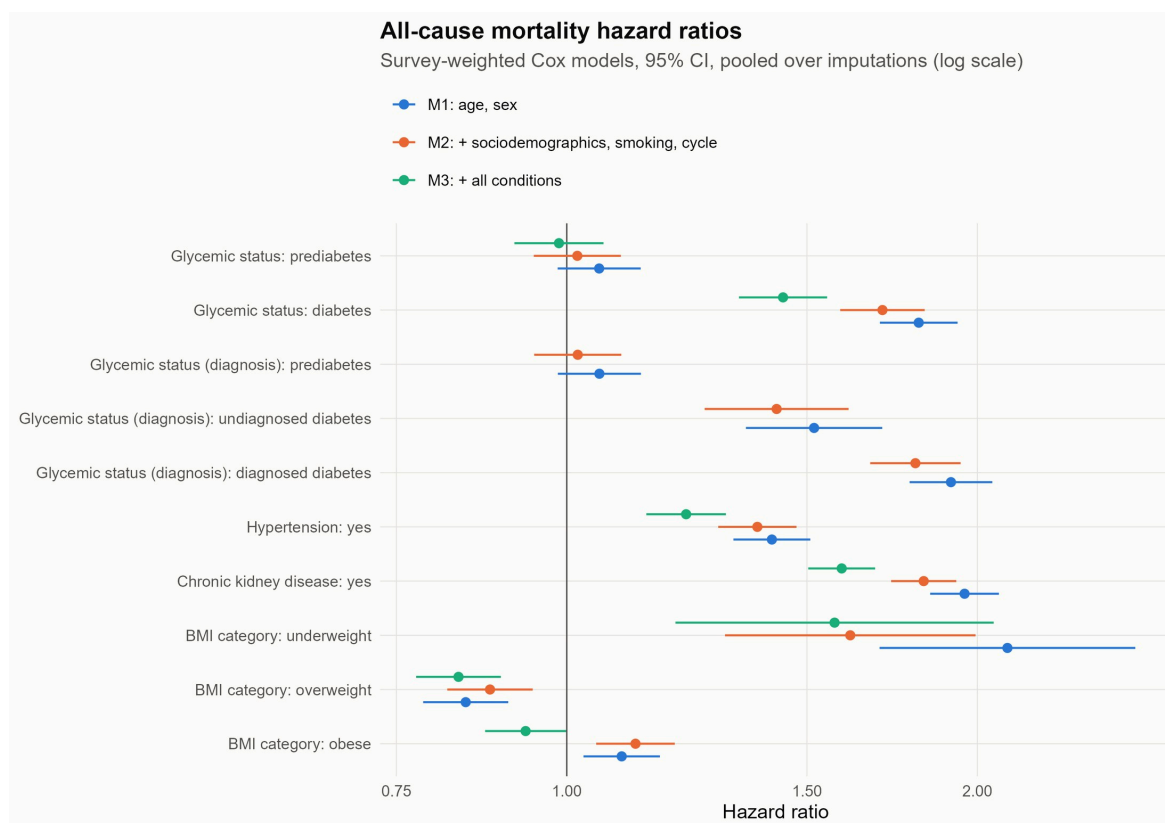


Figure 6.3 Hazard ratios with 95% CI by adjustment level (log scale).

Table 6.4 Full M3 model (age spline and cycle terms not shown).

Term	HR (95% CI)	p
Female (vs male)	0.68 (0.64 to 0.72)	<0.001
Non-Hispanic Black	0.96 (0.88 to 1.04)	0.296
Mexican American	0.67 (0.61 to 0.74)	<0.001
Other Hispanic	0.71 (0.58 to 0.85)	<0.001
Other/Multiracial	0.77 (0.66 to 0.90)	0.001
HS/GED (vs less)	0.92 (0.85 to 1.00)	0.055
More than HS (vs less)	0.87 (0.81 to 0.94)	<0.001
Income-to-poverty ratio, per unit	0.86 (0.84 to 0.88)	<0.001
Former smoker	1.22 (1.14 to 1.31)	<0.001
Current smoker	2.27 (2.09 to 2.46)	<0.001
Prediabetes	0.99 (0.92 to 1.06)	0.732
Diabetes	1.44 (1.34 to 1.55)	<0.001
Hypertension	1.22 (1.14 to 1.31)	<0.001
CKD	1.59 (1.50 to 1.68)	<0.001
Underweight	1.57 (1.20 to 2.06)	<0.001
Overweight	0.83 (0.78 to 0.89)	<0.001
Obese	0.93 (0.87 to 1.00)	0.046
Prior CVD	1.58 (1.48 to 1.69)	<0.001
Prior cancer	1.25 (1.17 to 1.34)	<0.001

Race reference: Non-Hispanic White. Smoking reference: never. p values from the pooled Wald test.

Current smoking had the largest hazard ratio of any modifiable factor in the model (HR 2.27, 95% CI 2.09 to 2.46). Each one-unit rise in the income-to-poverty ratio was linked to a 14% lower rate of death, even after adjusting for education and every condition. None of the survey cycle terms showed a steady pattern.

6.4 Dose-response

The EDA showed U and J shapes, so six continuous measures are modelled one at a time as natural cubic splines with M2 adjustment. Knots sit at the 35th and 65th percentiles, and boundary knots at the 5th and 95th. Each curve is the hazard ratio against a clinical reference value, over the 1st to 99th percentile of the measure.

Code 6.2 Building the spline term from percentiles and fitting it in every imputation.

r/06_survival_cox.R, lines 219-224

```
spline_curves <- pmap_dfr(spl, function(var, label, ref) {
  x <- dl[[var]]
  kn <- round(quantile(x, c(0.35, 0.65)), 3)
  bk <- round(quantile(x, c(0.05, 0.95)), 3)
  term <- sprintf("ns(%s, knots = c(%s), Boundary.knots = c(%s))", var, paste(kn, collapse = ", "), paste(bk, collapse = ", "))
  r <- cached(paste0("spline_", var), MIcombine(with(des, fun = function(d) svycoxph(cox_f(c(term, m2)), design = d))))
```

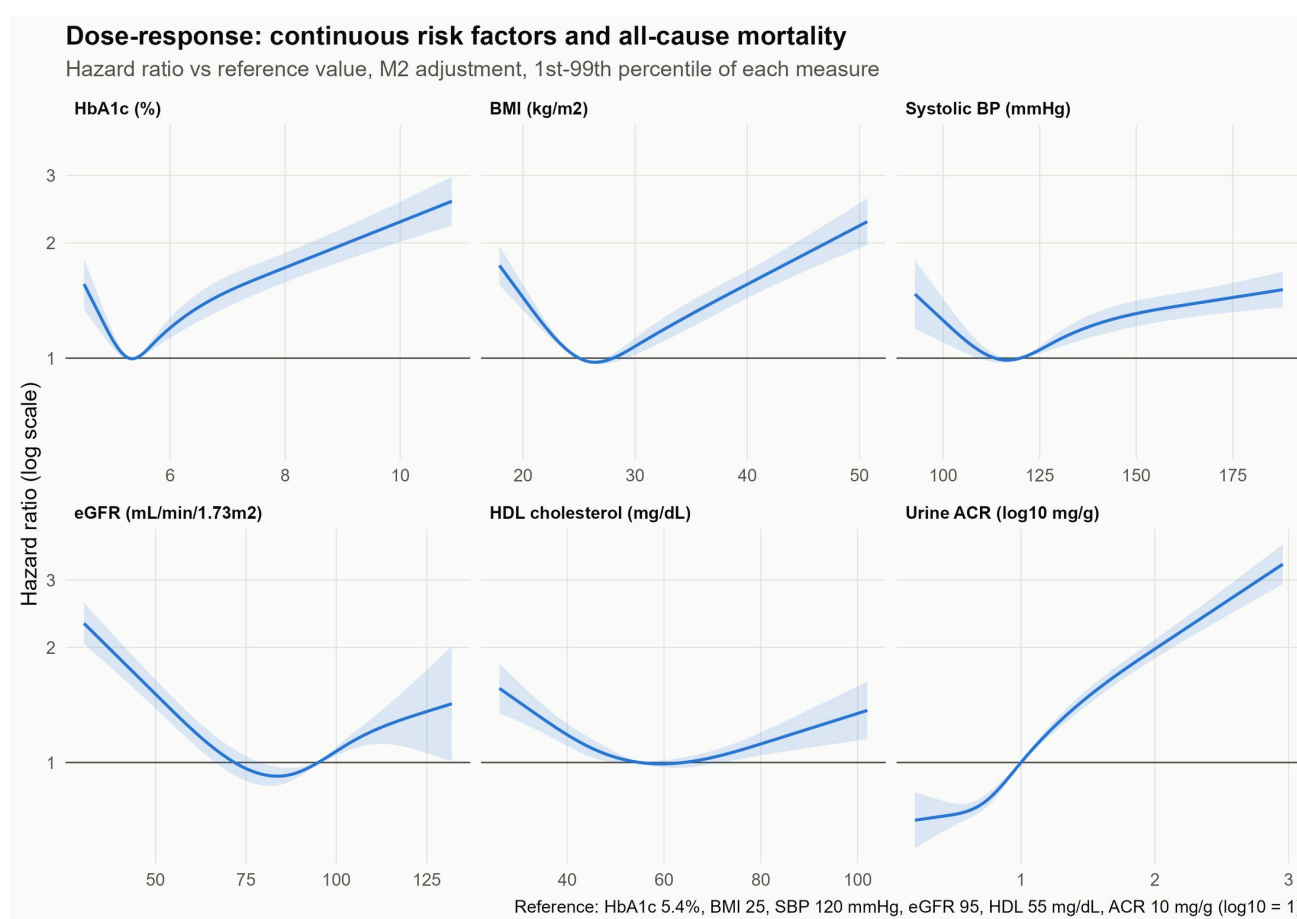


Figure 6.4 Hazard ratio across each measure (log scale), M2 adjustment. Reference: HbA1c 5.4%, BMI 25, SBP 120 mmHg, eGFR 95, HDL 55 mg/dL, ACR 10 mg/g.

Table 6.5 Where each curve is lowest, and the hazard ratio at the extremes.

Measure	Lowest risk at	HR at 1st pct	HR at 99th pct
BMI (kg/m ²)	26.5	1.75	2.27
HDL cholesterol (mg/dL)	59.2	1.56	1.37
HbA1c (%)	5.3	1.56	2.57
Systolic BP (mmHg)	116.6	1.47	1.51
Urine ACR (mg/g)	1.6	0.71	3.30
eGFR (mL/min/1.73m ²)	83.9	2.31	1.42

Risk was lowest at an HbA1c of 5.3%, a BMI of 26.5, systolic pressure of 116.6 mmHg and HDL of 59.2 mg/dL, and rose on both sides of each. An HbA1c of 4.5% (the 1st percentile) carried about the same risk as 7.0% (HR 1.56 vs 1.50). Low HbA1c is linked to higher mortality in other population studies too, and it may reflect frailty, anaemia or liver disease rather than good glucose control. ACR was different: risk rose steadily from the lowest values, up to an HR of 3.3 at the 99th percentile. The high-eGFR uptick from the EDA was still there with continuous age adjustment, though with a wide interval (HR 1.42, 95% CI 1.01 to 2.02 at the 99th percentile). A very high eGFR from creatinine often means low muscle mass rather than good kidneys, which is a known weakness of creatinine-based equations.

6.5 Proportional hazards

Table 6.6 Schoenfeld residual test, M3, imputation 1.

term	chisq	df	p
age (spline)	93.8	4	<0.001
sex	5.2	1	0.022
race_eth	4.6	4	0.327
education	2.8	2	0.241
pir	2.1	1	0.143
smoking	7.8	2	0.020
cycle	26.2	9	0.002
glycemic	7.0	2	0.030
hypertension	23.1	1	<0.001
ckd	4.8	1	0.028
bmi_cat	12.8	3	0.005
cvd_history	0.1	1	0.756
cancer	1.4	1	0.244
GLOBAL	187.6	32	<0.001

The Cox model assumes each hazard ratio stays the same over follow-up. The Schoenfeld residual test rejects this globally ($p < 0.001$), and for age, hypertension and BMI in particular. With 50,000 people, though, the test flags departures too small to matter. The more useful question is whether the hazard ratios change enough to alter the conclusions. The time-split models in Section 6.6 answer that directly.

6.6 Sensitivity analyses

Table 6.7 M3 hazard ratios under six alternative analyses.

Analysis	Prediabetes	Diabetes	Hypertension	CKD	Underweight	Overweight	Obese
Main: multiple imputation	0.99	1.44	1.22	1.59	1.57	0.83	0.93
Complete cases only (n = 39,607)	1.03	1.50	1.24	1.59	1.81	0.85	0.94
Excluding deaths in first 2 years	0.99	1.46	1.25	1.53	1.61	0.87	0.97
Excluding baseline CVD or cancer	1.01	1.55	1.28	1.60	1.60	0.83	0.99
Without survey-cycle adjustment	0.99	1.44	1.22	1.59	1.58	0.83	0.93
Follow-up years 0-10	0.96	1.39	1.20	1.72	1.61	0.81	0.90
Follow-up beyond 10 years	1.05	1.60	1.25	1.33	1.49	0.89	1.02

Full intervals are in outputs/R/06_survival_cox/06_sensitivity.csv and in the figure below.

The main results hold up. Diabetes stays between 1.39 and 1.60 in every analysis, prediabetes stays close to 1, and hypertension and CKD stay clearly above 1. Excluding deaths in the first two years moves the estimates only slightly, so the associations are not driven by people who were already very ill at the exam. Dropping survey cycle from the model changes no estimate by more than 0.01.

Two findings stand out. The complete-case estimates are a little further from 1 for diabetes (1.50) and underweight (1.81), which fits the selection problem shown in Section 4.4. And the time split shows the proportional hazards departure clearly. CKD had a stronger association in the first 10 years (1.72) than after (1.33), while diabetes went the other way (1.39 then 1.60). The M3 hazard ratios are therefore averages over the follow-up period. The direction never changes, so the conclusions do not either.

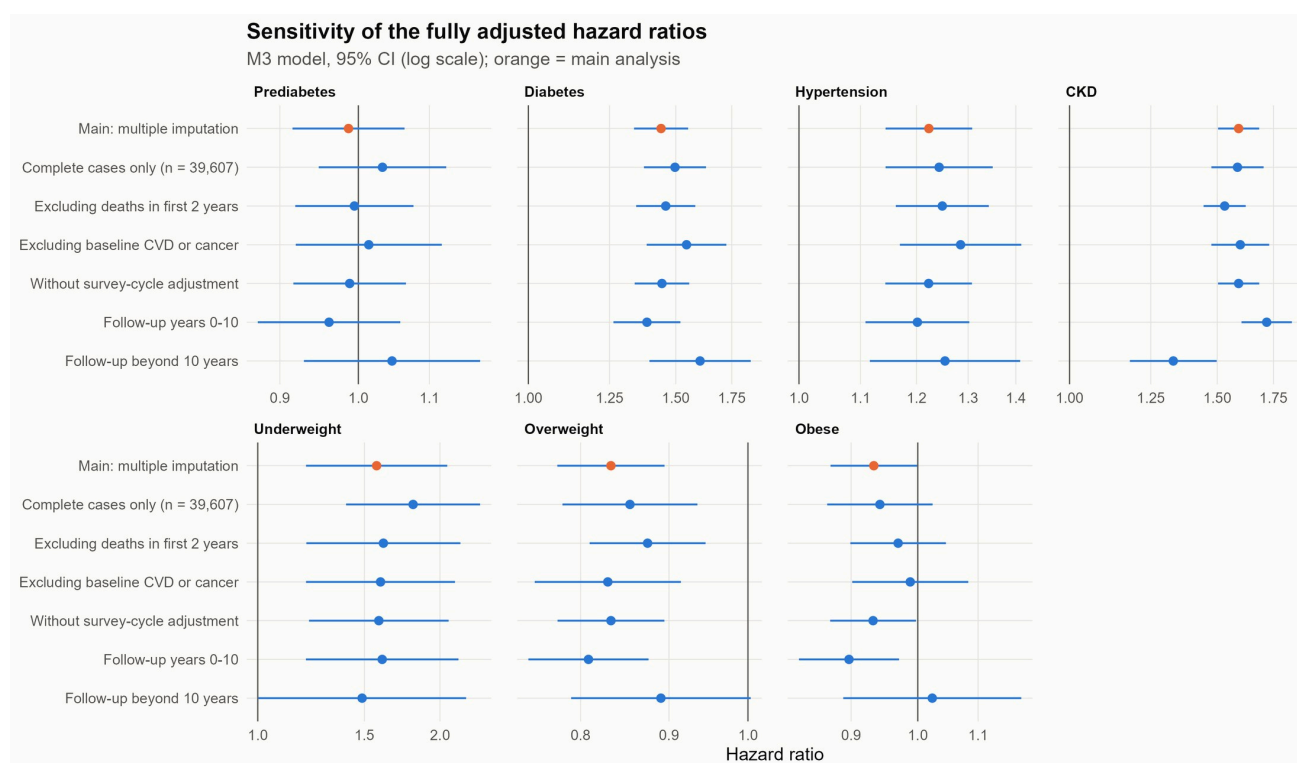


Figure 6.5 Sensitivity of the M3 hazard ratios, 95% CI (log scale). Orange marks the main analysis.

7 Competing risks and causes of death

Script 07. What people with each condition die of, the chance of dying from each cause when other causes compete, and cause-specific and Fine-Gray hazard ratios.

7.1 Cause-of-death mix

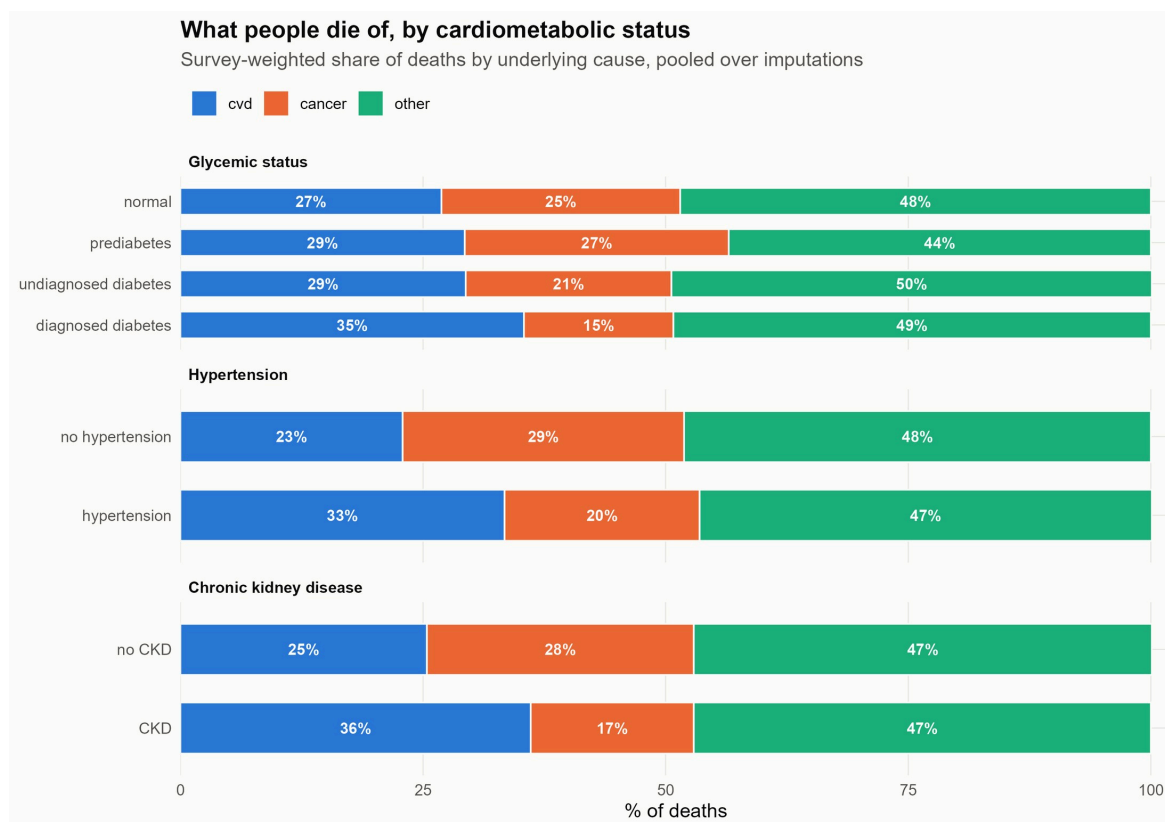


Figure 7.1 Survey-weighted share of deaths by underlying cause, pooled over imputations.

CVD made up 26.9% of deaths among adults with normal glucose and 35.4% among adults with diagnosed diabetes. The cancer share moved the other way, from 24.6% to 15.4%. A falling share does not mean a lower risk. It means that other causes took a larger part of the total. The same shift appears for hypertension and CKD.

Diabetes itself was the underlying cause of 12.6% of deaths in adults with diagnosed diabetes. It was mentioned anywhere on the death certificate for only 34.5% of them (95% CI 31.6 to 37.5), and for 16.3% of people whose diabetes was never diagnosed. Studies that count diabetes deaths from death certificates alone would miss about two thirds of the deaths in people who had diabetes.

7.2 Cause-specific death rates

Table 7.1 Age-standardized cause-specific deaths per 1,000 person-years by glycemic status.

Glycemic status	CVD	Cancer	Other causes
normal	3.29	2.72	5.55
prediabetes	3.47	3.30	5.37
undiagnosed diabetes	5.26	3.92	9.01
diagnosed diabetes	8.19	3.16	11.69

Diagnosed diabetes raised the age-standardized CVD death rate from 3.29 to 8.19 per 1,000 person-years, and other causes from 5.55 to 11.69. The cancer death rate hardly changed. So the lower cancer share in 7.1 comes from more deaths from other causes, not fewer cancer deaths.

7.3 Cumulative incidence

A common mistake is to estimate the risk of CVD death as 1 minus a Kaplan-Meier curve with other deaths censored. That treats people who died of cancer as if they could still die of CVD later, and it overstates the risk. The Aalen-Johansen estimator accounts for competing deaths. Here it is fitted with `survfit` on a multi-state outcome, weighted by the survey weight, and averaged over imputations.

Code 7.1 Aalen-Johansen cumulative incidence alongside the naive 1 minus Kaplan-Meier estimate. `r/07_competing_risks.R`, lines 151-157

```
grid <- seq(0, 18, by = 0.25)
cif <- map_dfr(names(groups), function(v) map_dfr(split(long, long$.imp), function(d) {
  s <- summary(survfit(reformulate(v, "Surv(time_y, cause_f)"), data = d, weights = wt_mec), times = grid, extend = TRUE)
  km <- map_dfr(names(causes), function(k) {
    sk <- summary(survfit(reformulate(v, sprintf("Surv(time_y, d_%s)", k))), data = d, weights = wt_mec), times = grid,
    extend = TRUE)
    tibble(level = sub(".*=", "", sk$strata), time = sk$time, cause = causes[[k]], naive = 1 - sk$surv)
  })
})
```

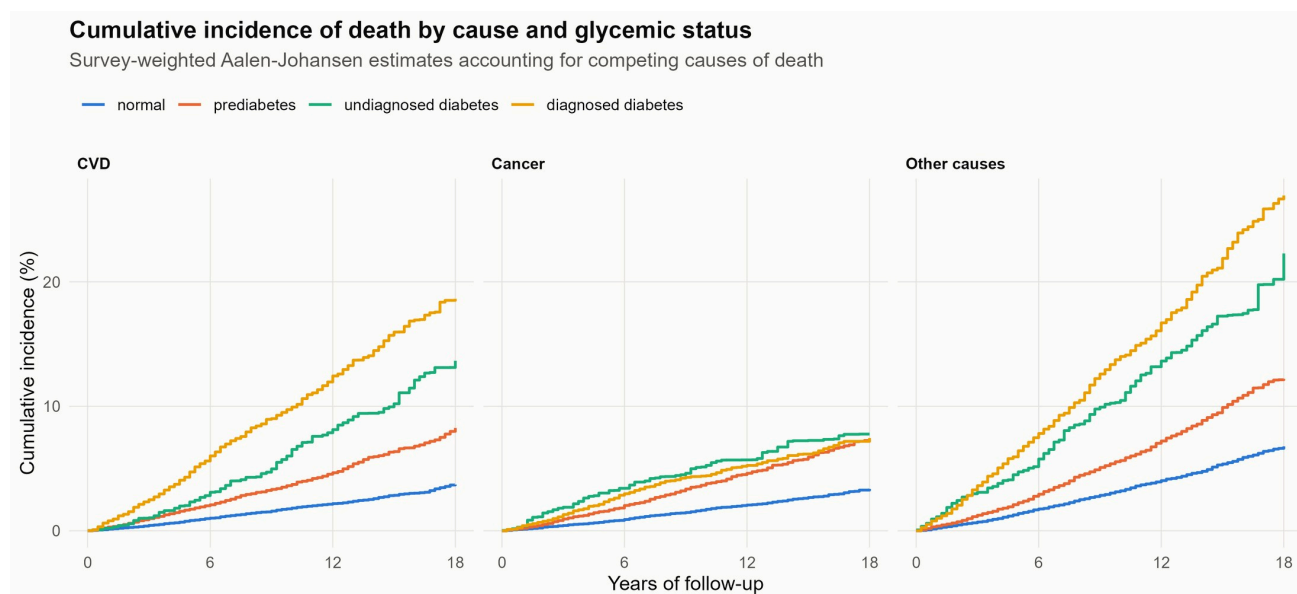


Figure 7.2 Cumulative incidence of death by cause and glycemic status (weighted Aalen-Johansen).

Table 7.2 15-year cumulative incidence of CVD death: Aalen-Johansen vs the naive estimate.

Group	Level	Aalen-Johansen %	1 - KM %	Overstated by %
Glycemic status	normal	2.8	3.0	4.1
	prediabetes	6.4	6.9	8.9
	undiagnosed diabetes	10.2	11.9	16.7
	diagnosed diabetes	16.0	19.0	19.3
Hypertension	no hypertension	2.0	2.1	3.8
	hypertension	11.6	13.2	14.3
Chronic kidney disease	no CKD	3.0	3.2	5.0
	CKD	17.6	21.4	21.4

Over 15 years, 16.0% of adults with diagnosed diabetes died of CVD, against 2.8% with normal glucose. The naive method overstated CVD risk by 19.3% in diagnosed diabetes and 21.4% in CKD. The error is largest exactly where competing deaths are most common, which is where the estimate matters most. For cancer death in CKD it reached 28.5%.

7.4 Cause-specific hazard ratios

Cause-specific Cox models treat other causes as censoring and use the same M3 adjustment as Section 6. They answer the question "how much faster does this cause kill people with the condition?"

Table 7.3 Cause-specific hazard ratios (95% CI), M3 adjustment (diagnosis split: M2).

Exposure	All causes	CVD	Cancer	Other causes
Prediabetes	0.99 (0.92 to 1.06)	0.97 (0.86 to 1.10)	1.10 (0.96 to 1.26)	0.94 (0.84 to 1.04)
Diabetes	1.44 (1.34 to 1.55)	1.50 (1.33 to 1.69)	1.03 (0.86 to 1.23)	1.62 (1.45 to 1.81)
Undiagnosed diabetes (M2)	1.43 (1.26 to 1.61)	1.45 (1.17 to 1.80)	1.21 (0.93 to 1.56)	1.53 (1.28 to 1.82)
Diagnosed diabetes (M2)	1.80 (1.67 to 1.94)	2.26 (2.00 to 2.54)	1.10 (0.91 to 1.33)	1.91 (1.71 to 2.13)
Hypertension	1.22 (1.14 to 1.31)	1.46 (1.28 to 1.68)	0.98 (0.85 to 1.12)	1.26 (1.14 to 1.38)
CKD	1.59 (1.50 to 1.68)	1.85 (1.65 to 2.08)	1.27 (1.11 to 1.45)	1.59 (1.45 to 1.75)
Underweight	1.57 (1.20 to 2.06)	1.41 (0.85 to 2.35)	1.83 (1.11 to 3.01)	1.57 (1.19 to 2.06)
Overweight	0.83 (0.78 to 0.89)	0.88 (0.77 to 1.02)	0.97 (0.82 to 1.15)	0.75 (0.67 to 0.83)
Obese	0.93 (0.87 to 1.00)	1.09 (0.95 to 1.25)	1.07 (0.91 to 1.26)	0.80 (0.72 to 0.89)
Current smoking	2.27 (2.09 to 2.46)	1.87 (1.63 to 2.15)	2.98 (2.53 to 3.51)	2.24 (1.96 to 2.56)

Diabetes was linked to CVD death (HR 1.50, 95% CI 1.33 to 1.69) and to death from other causes (HR 1.62, 95% CI 1.45 to 1.81), but not to cancer death (HR 1.03, 95% CI 0.86 to 1.23). Diagnosed diabetes more than doubled the CVD death rate in M2 (HR 2.26, 95% CI 2.00 to 2.54). Hypertension was a CVD risk factor (HR 1.46) with no link to cancer. CKD was linked to all three causes, most strongly to CVD (1.85). Current smoking had its largest HR for cancer (2.98), as expected.

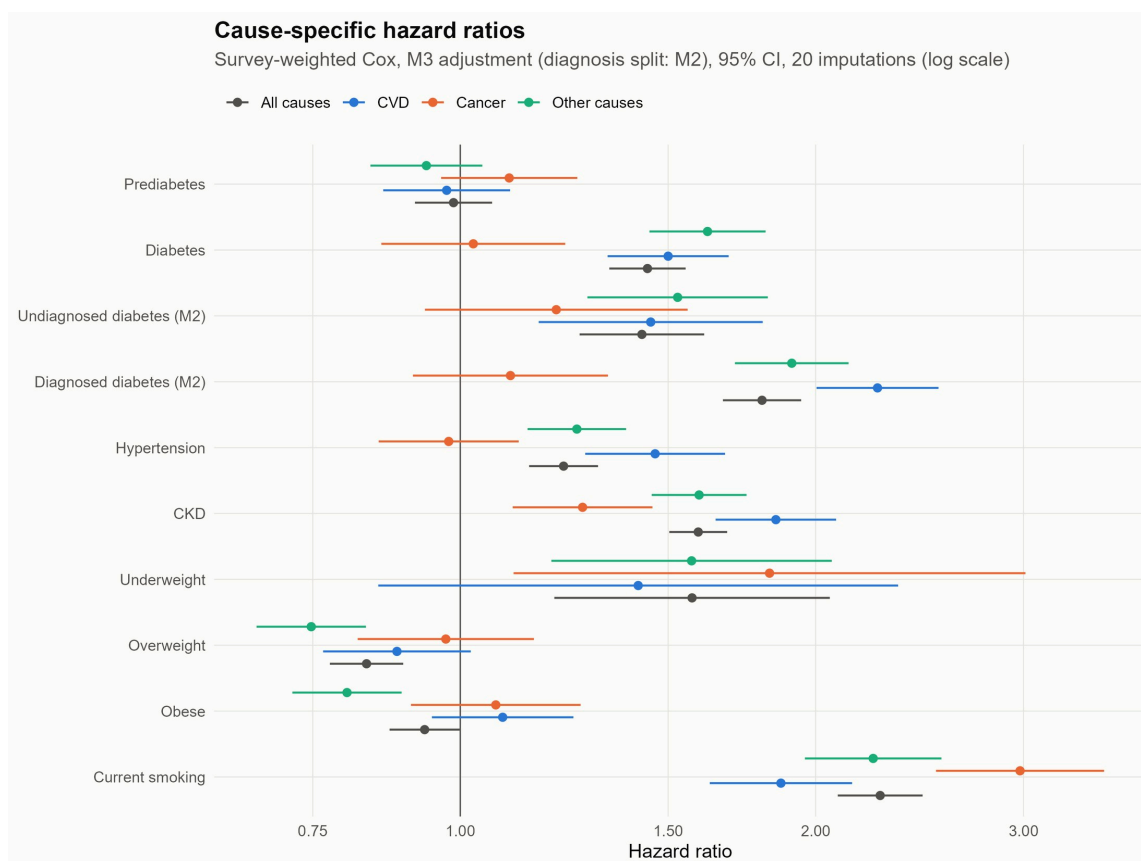


Figure 7.3 Cause-specific hazard ratios, 95% CI (log scale).

7.5 Fine-Gray subdistribution hazards

The Fine-Gray model answers a different question: how much more likely is death from this cause in the real world, where other causes also kill. It keeps people who died of other causes in the risk set with a decreasing weight. `finegray()` expands the data a lot, so follow-up is rounded up to quarter-years (890,000 rows became 350,000, and the HRs did not change at 3 decimals) and 5 of the 20 imputations are used. Rubin's rules remain valid with 5.

Code 7.2 Fine-Gray model: expanded data, then a survey Cox fit weighted by survey weight times the Fine-Gray weight. r/07_competing_risks.R, lines 247-249

```
fg_fit <- function(i, k) {
  fgd <- finegray(Surv(time_y, cause_f) ~ ., data = filter(long, .imp == i) %>% mutate(time_y = ceiling(time_months / 3)
/ 4) %>% select(all_of(fg_vars)), etype = k)
  fit <- svycoxph(cox_f(m3, "fgstart", fgstop", "fgstatus"), design = svydesign(ids = ~psu, strata = ~strata, weights =
~I(wt_mec * fgwt)), nest = TRUE, data = fgd))
```

Table 7.4 Cause-specific vs Fine-Gray hazard ratios (95% CI), M3 adjustment.

Cause	Exposure	Cause-specific HR	Subdistribution HR (Fine-Gray)
CVD	Current smoking	1.87 (1.63-2.15)	1.57 (1.37-1.80)
	Diabetes	1.50 (1.33-1.69)	1.38 (1.21-1.57)
	Hypertension	1.46 (1.28-1.68)	1.46 (1.27-1.67)
	CKD	1.85 (1.65-2.08)	1.64 (1.45-1.85)
	Underweight	1.41 (0.85-2.35)	1.20 (0.75-1.92)
	Obese	1.09 (0.95-1.25)	1.15 (1.00-1.31)
Cancer	Current smoking	2.98 (2.53-3.51)	2.73 (2.32-3.20)
	Diabetes	1.03 (0.86-1.23)	0.96 (0.80-1.14)
	Hypertension	0.98 (0.85-1.12)	0.97 (0.85-1.11)
	CKD	1.27 (1.11-1.45)	1.10 (0.95-1.26)
	Underweight	1.83 (1.11-3.01)	1.64 (1.02-2.66)
	Obese	1.07 (0.91-1.26)	1.13 (0.96-1.33)

Most subdistribution HRs are a little closer to 1 than the cause-specific ones. Diabetes and CVD death is 1.38 (1.21-1.57) against 1.50 (1.33-1.69). This is expected: people with diabetes also die sooner of other causes, so fewer of them live long enough to die of CVD. CKD and cancer death loses significance under Fine-Gray (1.10 (0.95-1.26)). The cause-specific HR is the better measure of how a condition acts on a cause. The Fine-Gray HR is the better measure for predicting who will actually die of it.

8 Causal effects and attributable burden

Script 08. Inverse probability weighting for the total effect of diabetes, hypertension and CKD, a check of covariate balance, E-values for unmeasured confounding, and the share of deaths attributable to each risk factor.

This section uses causal methods on observational data. They estimate what would happen under clear assumptions: no unmeasured confounding, correct models and enough overlap between groups. The results are best read as "what the effect would be if these assumptions hold", and Section 8.5 tests how strong a hidden confounder would need to be to change them.

8.1 Confounders and mediators

The M3 model in Section 6 adjusts every condition for every other one. For diabetes that includes hypertension and CKD, which diabetes itself causes. Adjusting for a mediator removes part of the effect that runs through it. For the total effect, each exposure gets its own adjustment set: common causes of the exposure and death, and nothing downstream.

Table 8.1 Adjustment set for each exposure.

Exposure	Adjusted for (confounders)	Not adjusted for (mediators)
Diabetes	Age (spline), sex, race/ethnicity, education, income, smoking, cycle, BMI (spline)	Hypertension, CKD
Hypertension	Same, plus diabetes	CKD
CKD	Same, plus diabetes and hypertension	None in the model

8.2 Propensity scores and weights

For each exposure and each imputation, a survey-weighted logistic model gives the propensity score, the probability of having the exposure given the confounders. The stabilized weight is $P(A = a)$ divided by $P(A = a | L)$, multiplied by the survey weight. Weighting by it creates a population in which the exposure is unrelated to the measured confounders.

Code 8.1 Propensity score and stabilized inverse probability weight, per imputation.

r/08_causal_attributable.R, lines 74-80

```
add_ipw <- function(d, e) {
  f <- reformulate(spec[[e]]$conf, e)
  ps <- fitted(svyglm(f, svydesign(ids = ~psu, strata = ~strata, weights = ~wt_mec, nest = TRUE, data = d), family =
    quasibinomial()))
  pa <- wmean(d[[e]], d$wt_mec)
  sw <- if_else(d[[e]] == 1, pa / ps, (1 - pa) / (1 - ps))
  mutate(d, ps = ps, sw = sw, w_ipw = wt_mec * sw)
}
```

Table 8.2 Weight summary, imputation 1. Stabilized weights should average close to 1.

Exposure	Exposed %	PS min	PS max	SW mean	SW 1st pct	SW 99th pct	SW max
Diabetes	12.4	0.001	0.987	1.005	0.20	2.49	69.4
Hypertension	32.0	0.006	0.991	1.008	0.35	5.44	40.3
Chronic kidney disease	13.6	0.021	0.935	1.003	0.18	2.96	8.5

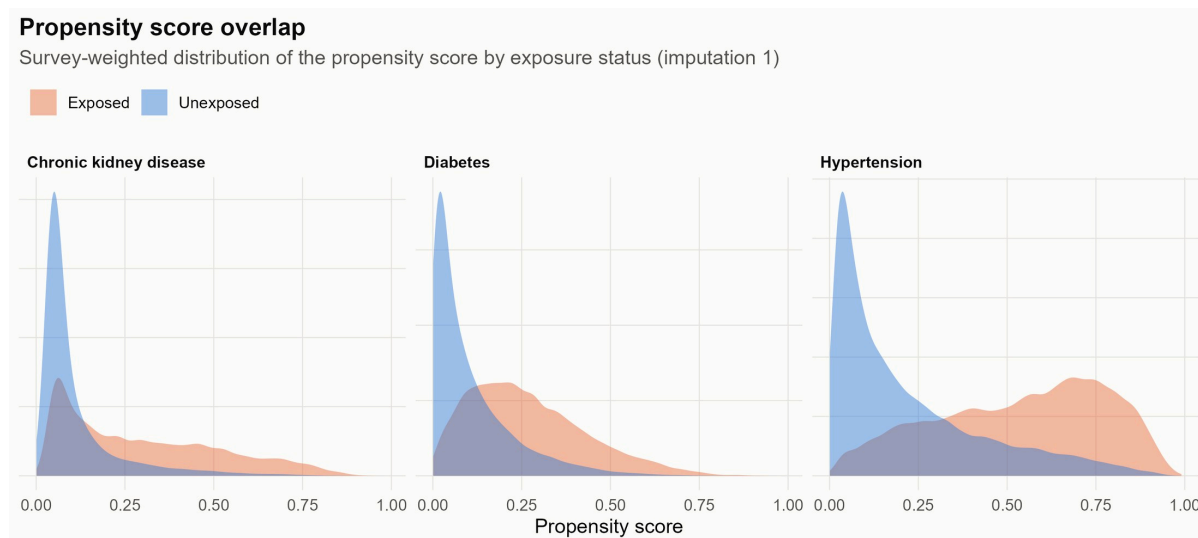


Figure 8.1 Weighted propensity score distribution by exposure status (imputation 1).

The mean stabilized weights are 1.005, 1.008, 1.003, as they should be. Overlap is good for CKD and hypertension. For diabetes, the unexposed group has a very long spike near zero, and the largest weight is 69.4. Weights are often truncated at the 1st and 99th percentiles to limit such values, and that was tried. It made balance worse: after 1% truncation the largest standardized difference for diabetes went from 0.077 to 0.324, mostly in age, and the HR moved from 1.83 to 2.28. The main analysis therefore uses untruncated weights.

Table 8.3 Weight truncation sensitivity.

Exposure	Truncated per tail	HR (95% CI)	Max SMD
Diabetes	0%	1.83 (1.58-2.13)	0.077
	0.1%	1.98 (1.75-2.24)	0.093
	0.5%	2.19 (1.98-2.42)	0.261
	1%	2.28 (2.07-2.50)	0.324
Hypertension	0%	1.17 (1.07-1.28)	0.114
	0.1%	1.20 (1.10-1.31)	0.106
	0.5%	1.26 (1.16-1.37)	0.088
	1%	1.32 (1.23-1.43)	0.137
Chronic kidney disease	0%	1.59 (1.47-1.72)	0.106
	0.1%	1.60 (1.47-1.73)	0.107
	0.5%	1.63 (1.50-1.76)	0.113
	1%	1.67 (1.55-1.81)	0.116

8.3 Covariate balance

Table 8.4 Standardized mean differences before and after weighting.

Exposure	Max SMD before	Over 0.1 before	Max SMD after	Over 0.1 after
Chronic kidney disease	0.895	9	0.106	1
Diabetes	0.881	7	0.077	0
Hypertension	1.255	8	0.114	1

Before weighting, age differed sharply between exposed and unexposed people (SMD 1.25 for hypertension). After weighting, every covariate for diabetes was under 0.1. For hypertension and CKD one covariate stayed just above 0.1, and in both cases it was sex (-0.114 and 0.106). That is a small residual imbalance, and the E-values in 8.5 give a sense of how much bias it could carry.

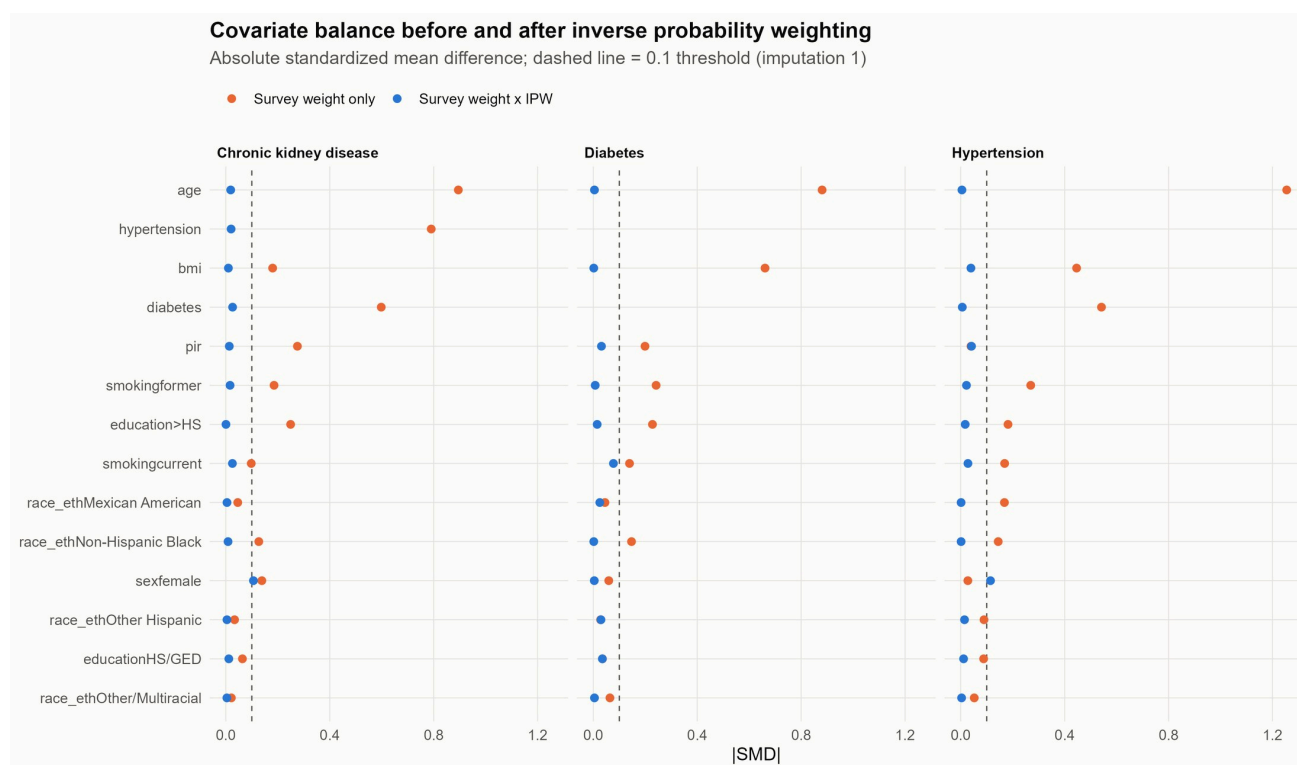


Figure 8.2 Absolute standardized mean difference before and after weighting (imputation 1). Dashed line at 0.1.

8.4 Marginal hazard ratios and adjusted survival

Table 8.5 IPW marginal hazard ratio (total effect) vs the conditional M3 hazard ratio, and the IPW-adjusted 10-year risk of death.

Exposure	IPW marginal HR	Conditional HR (06, M3)	10-y risk, unexposed %	10-y risk, exposed %	Difference (points)
Diabetes	1.83 (1.58-2.13)	1.44 (1.34-1.55)	9.9	16.2	6.3
Hypertension	1.17 (1.07-1.28)	1.22 (1.14-1.31)	10.7	12.7	2.0
Chronic kidney disease	1.59 (1.47-1.72)	1.59 (1.50-1.68)	9.4	16.0	6.6

The marginal HR for diabetes was 1.83 (95% CI 1.58 to 2.13), well above the conditional 1.44. The gap is the part of the diabetes effect that runs through hypertension and CKD, which M3 adjusts away. In absolute terms, the weighted 10-year risk of death was 16.2% with diabetes and 9.9% without, a difference of 6.3 percentage points. For CKD the marginal and conditional HRs were the same (6.6 points of extra 10-year risk), which makes sense because nothing downstream of CKD is in its model. Hypertension had the smallest effect, 2.0 points over 10 years.

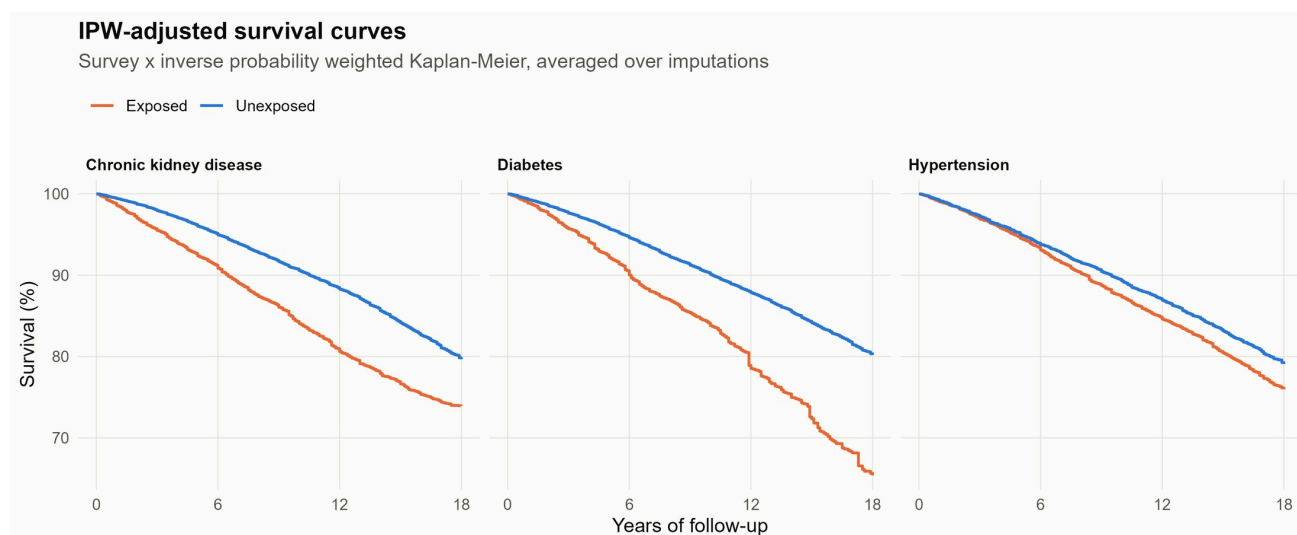


Figure 8.3 Survey and IPW-weighted survival curves, averaged over imputations.

8.5 E-values

An E-value is the minimum strength of association, on the risk ratio scale, that an unmeasured confounder would need with both the exposure and death to fully explain away an estimate. Deaths are common here (16%), so hazard ratios are first converted to risk ratios with VanderWeele's approximation.

Table 8.6 E-values for the main estimates.

Estimate	HR (95% CI)	E-value (point)	E-value (CI limit)
Diabetes (IPW marginal)	1.83 (1.58 to 2.13)	2.41	2.08
Hypertension (IPW marginal)	1.17 (1.07 to 1.28)	1.48	1.28
Chronic kidney disease (IPW marginal)	1.59 (1.47 to 1.72)	2.10	1.93
Current smoking (M3 conditional)	2.27 (2.09 to 2.46)	2.91	2.71
Diabetes (M3 conditional)	1.44 (1.34 to 1.55)	1.90	1.75
Hypertension (M3 conditional)	1.22 (1.14 to 1.31)	1.56	1.42
Chronic kidney disease (M3 conditional)	1.59 (1.50 to 1.68)	2.10	1.98
Underweight (M3 conditional)	1.57 (1.20 to 2.06)	2.08	1.53

To explain away the diabetes effect, a hidden confounder would need a risk ratio of 2.41 with both diabetes and death, over and above everything already adjusted for. To move the confidence limit to 1 it would need 2.08. Physical activity and diet are the obvious missing confounders, and a factor that strong beyond BMI, income, education and smoking seems unlikely. The hypertension estimate is more fragile, with an E-value of 1.48 for the point estimate and 1.28 for the confidence limit.

8.6 Population attributable fractions

The population attributable fraction (PAF) is the share of all deaths that would not have happened without the exposure, if the association were causal. It uses Miettinen's formula, $PAF = p_c \times (HR - 1) / HR$, where p_c is the weighted share of deaths that occurred in exposed people and HR is the fully adjusted M3 hazard ratio (cause-specific M3 for CVD deaths). The M3 HRs are used on purpose. Each condition is adjusted for the others, so the fractions do not count the same deaths twice through shared pathways. They still overlap somewhat and should not be added up.

Code 8.2 PAF with Miettinen's formula, all-cause and CVD, with the interval taken from the HR `r/08_causal_attributable.R`, lines 238-246 interval.

```
paf <- pmap_dfr(paf_exposures, function(term, label, indicator) map_dfr(list(`All-cause` = list(m3, "death"), CVD =
list(cs_cvd, "d_cvd")), function(x) {
  h <- filter(x[[1]], term == !!term)
  pc <- mean(map_dbl(imps, function(d) {
    dd <- filter(mutate(d, d_cvd = as.numeric(cause == "cvd")), .data[[2]] == 1)
    wmean(as.numeric(eval(parse(text = indicator), dd)), dd$wt_mec)
  })))
  tibble(exposure = label, hr = exp(h$est), hr_lcl = exp(h$lcl), hr_ucl = exp(h$ucl), pct_deaths_exposed = 100 * pc) %>%
    mutate(paf_pct = pct_deaths_exposed * (hr - 1) / hr, paf_lcl = pct_deaths_exposed * (hr_lcl - 1) / hr_lcl, paf_ucl =
    pct_deaths_exposed * (hr_ucl - 1) / hr_ucl)
  }, .id = "outcome")) %>%
```

Table 8.7 Share of deaths attributable to each risk factor in US adults aged 20+.

Risk factor	Deaths in exposed %	All-cause PAF %	CVD PAF %
Chronic kidney disease	39.5	14.7 (13.2 to 16.1)	22.2 (18.9 to 25.1)
Current smoking	23.5	13.1 (12.2 to 13.9)	8.7 (7.2 to 10.0)
Hypertension	64.5	11.8 (8.1 to 15.2)	22.9 (15.7 to 29.3)
Diabetes	27.2	8.3 (6.9 to 9.7)	10.4 (7.8 to 12.8)
Former smoking	37.1	6.7 (4.4 to 8.8)	0.2 (-4.3 to 4.2)
Underweight	2.5	0.9 (0.4 to 1.3)	0.5 (-0.3 to 1.0)

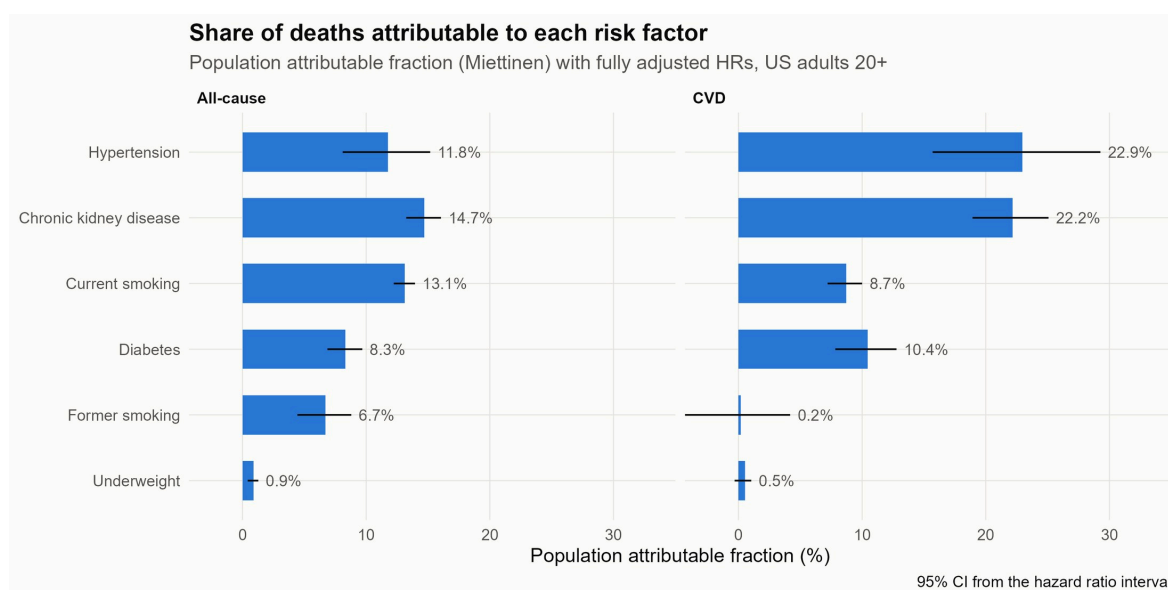


Figure 8.4 Population attributable fractions, all-cause and CVD mortality, with 95% CI.

CKD accounted for the largest share of all deaths (14.7%), ahead of current smoking (13.1%), hypertension (11.8%) and diabetes (8.3%). For CVD deaths, hypertension (22.9%) and CKD (22.2%) each accounted for over a fifth. Hypertension has a modest hazard ratio but a large PAF, because 64% of the people who died had it. Diabetes is the reverse: a stronger hazard ratio but fewer people. CKD is less often talked about than diabetes or smoking, yet in this data it carries the largest attributable share.

9 Predicting 10-year risk of death

Script 09, part A. Three Cox models developed on the early cycles and validated on later cycles, with calibration, decision curves and risk tiers.

9.1 Temporal split

A random split would let the model see people from the same survey years it is tested on. A temporal split is a harder and more honest test: the models are developed on 1999-2006 and validated on 2007-2010, as if they had been built in 2007 and then used on new people. The 2011-2018 cycles do not have 10 years of follow-up, so they are not used for validation. Survey cycle is never a predictor, because a model used today cannot know it. Survey weights are used throughout, so performance refers to the US adult population.

Table 9.1 Development and validation samples (imputation 1).

Sample	Cycles	n	deaths	deaths within 10 y	min follow-up, y (alive)
development	1999-2000 to 2005-2006	17,869	5,021	2,959	12.8
validation	2007-2008 to 2009-2010	11,621	1,860	1,652	8.8
later cycles	2011-2012 to 2017-2018	21,329	1,398	1,398	0.9

9.2 Three models

Table 9.2 Candidate models.

Model	Predictors
A: Basic	Age (spline), sex, race/ethnicity, smoking
B: Conditions	A plus glycemic status with diagnosis, hypertension, CKD, BMI group, prior CVD, prior cancer
C: Full	A plus education, income, splines for BMI, SBP, HbA1c, eGFR, log ACR and HDL, total cholesterol, BP medication, diagnosed diabetes, prior CVD, prior cancer

Each model is a weighted Cox model fitted in all 20 imputations. The coefficients and the 10-year baseline cumulative hazard are averaged across imputations, and 10-year risk is $1 - \exp(-H_0(10) \times \exp(\text{linear predictor}))$. Model B uses only yes/no conditions a clinic would know. Model C adds the lab values themselves.

Code 9.1 Pooling a Cox model over imputations and predicting 10-year risk.

r/09_risk_prediction_powerbi.R, lines 95-109

```
fit_pooled <- function(terms, data_list) {
  f <- reformulate(terms, "Surv(time_y, death)")
  fits <- map(data_list, function(d) {
    fit <- coxph(f, data = d, weights = w, model = TRUE)
    bh <- basehaz(fit, centered = FALSE)
    list(beta = coef(fit), H0 = max(c(0, bh$hazard[bh$time <= horizon])), tt = delete.response(terms(fit)), xlevels = fit$xlevels)
  })
  list(terms = terms, tt = fits[[1]]$tt, xlevels = fits[[1]]$xlevels,
       beta = rowMeans(sapply(fits, function(x) x$beta)), H0 = mean(map_dbl(fits, "H0")))
}

predict_risk <- function(model, d) {
  X <- model.matrix(model$tt, model.frame(model$tt, d, xlev = model$xlevels))[, -1, drop = FALSE]
  lp <- drop(X[, names(model$beta)] %*% model$beta)
  tibble(lp = lp, risk = 1 - exp(-model$H0 * exp(lp)))
}
```

9.3 Validation

Five measures describe performance in the 2007-2010 validation sample. The C-index measures ranking: the chance that of two people, the one who died first had the higher predicted risk. The IPCW Brier score measures prediction error at 10 years with weights for censoring, and the scaled Brier compares it with predicting the average risk for everyone. The observed to expected (O/E) ratio and calibration slope check whether the predicted risks are the right size.

Table 9.3 Temporal validation, 2007-2010, averaged over 20 imputations.

Model	C-index	Brier	Scaled Brier	Observed %	Predicted %	O/E	Cal. slope
A: Basic	0.850	0.0678	0.270	10.4	10.3	1.003	0.988
B: Conditions	0.862	0.0653	0.297	10.4	10.4	0.995	0.951
C: Full	0.872	0.0622	0.330	10.4	10.2	1.014	0.983

All three models rank people well. The basic model alone reaches a C-index of 0.850, because age explains most of the ranking. Adding the conditions raises it to 0.862, and adding the lab values to 0.872. The scaled Brier score improves in the same order, from 0.270 to 0.330. All three are well calibrated on average: the observed 10-year risk was 10.4% and model C predicted 10.2% (O/E 1.014), with calibration slopes between 0.95 and 0.99.

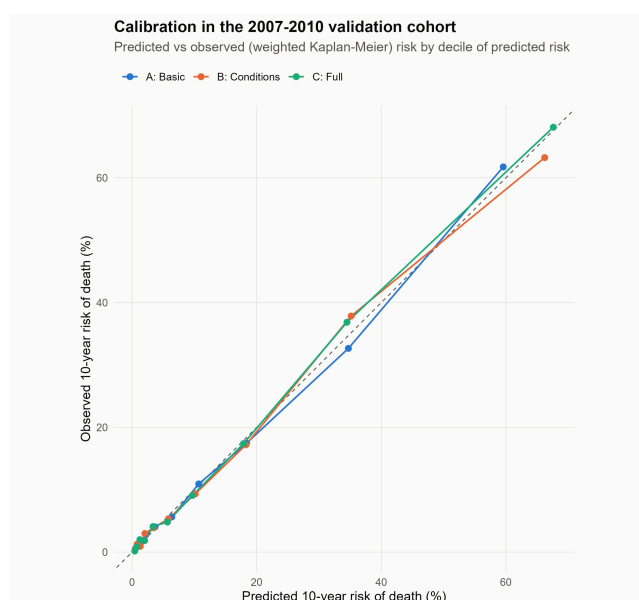


Figure 9.1 Predicted vs observed 10-year risk by decile of predicted risk, validation sample.

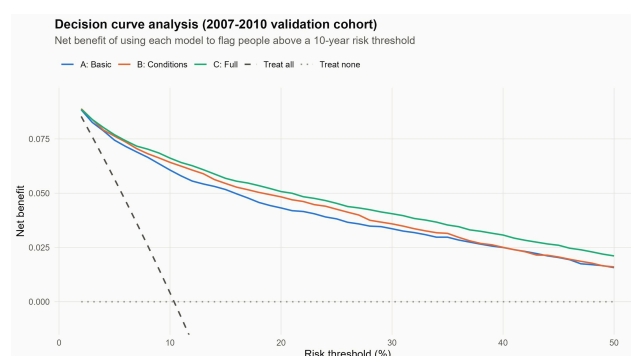


Figure 9.2 Decision curve: net benefit of flagging people above each risk threshold.

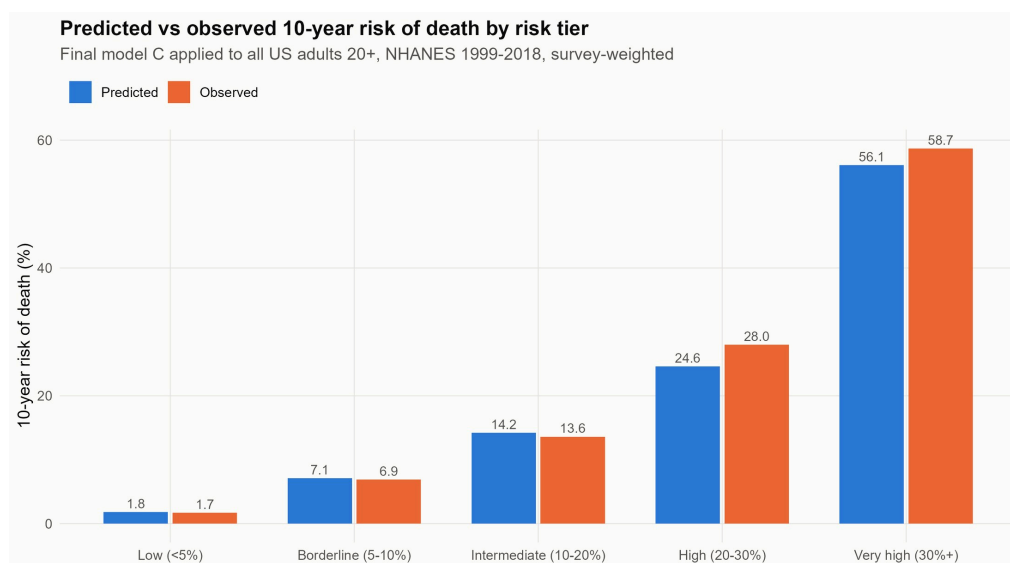
Decision curve analysis asks whether using a model to decide who gets follow-up does more good than harm, compared with flagging everyone or no one. At a 10% risk threshold, model C had a net benefit of 0.066, against 0.004 for flagging everyone. In plain terms, using model C at that threshold is equal to finding about 6.6 extra deaths per 100 people screened with no false alarms. Model C is above or level with the other two at almost every threshold, although the gaps are small.

9.4 Final model and risk tiers

After validation, models B and C are refitted on all ten cycles, which is standard practice once the method has been tested. Model C is used to place every adult in a risk tier. Model B, which only needs yes/no answers, drives the what-if calculator in the dashboard.

Table 9.4 Final model B hazard ratios (all cycles, age spline not shown). Point estimates averaged over imputations.

Term	HR
Female (vs male)	0.73
Non-Hispanic Black	1.09
Mexican American	0.85
Other Hispanic	0.87
Other/Multiracial	0.81
Former smoker	1.23
Current smoker	2.56
Prediabetes	1.00
Undiagnosed diabetes	1.38
Diagnosed diabetes	1.54
Hypertension	1.25
CKD	1.62
Underweight	1.73
Overweight	0.84
Obese	0.95
Prior CVD	1.64
Prior cancer	1.21

**Figure 9.3** Predicted vs observed 10-year risk by tier (final model C, all cycles, survey-weighted).**Table 9.5** Risk tiers in US adults (final model C, survey-weighted).

Risk tier	% of adults	Mean age	Predicted %	Observed %	Diabetes %	HTN %	CKD %
Low (<5%)	61.4	36.9	1.8	1.7	4.3	14.3	4.8
Borderline (5-10%)	13.3	55.1	7.1	6.9	16.3	44.7	10.7
Intermediate (10-20%)	10.0	61.5	14.2	13.6	23.7	57.6	19.0
High (20-30%)	4.4	67.0	24.6	28.0	30.8	67.4	28.8
Very high (30%+)	11.0	75.1	56.1	58.7	35.3	77.7	55.4

61.4% of US adults fell in the low tier, with a predicted 10-year risk of 1.8% and an observed risk of 1.7%. The very high tier held 11.0% of adults, with a mean age of 75.1 and an observed 10-year risk of 58.7%. Predicted and observed risks agree closely in every tier except the high tier, where the model under-predicted a little (24.6% vs 28.0%). The very high tier is mostly older adults, and 55.4% of them had CKD.

For the what-if page, model B predicts 10-year risk for every combination of age (30 to 80 in steps of 10), sex, race and ethnicity, smoking, glycemic status, hypertension, CKD and BMI group, with no prior CVD or cancer: 11,520 profiles in all. Power BI only looks these up, so the dashboard shows exactly what the R model predicts.

10 Power BI dashboard

Script 09, part B, and the Power BI report. How the R results become a dashboard without any number being recomputed in DAX.

10.1 Data model

The last part of script 09 writes the dashboard data model straight to `outputs/powerbi`: 42 tables (7 dimensions, 1 fact table, 28 statistical result tables and 6 model result tables), a list of 35 relationships and a table dictionary. Every dimension has one key column with the same name in every table that uses it, and a sort column. Before writing, the script checks every relationship for keys that have no match in their dimension and stops if it finds any.

Code 10.1 Referential integrity check before the Power BI tables are written.

`r/09_risk_prediction_powerbi.R`, lines 430-435

```
unmatched_keys = pmap_int(list(from_table, from_column, to_table, to_column), function(ft, fc, tt, tc)
length(setdiff(unique(tables[[ft]][[fc]]), tables[[tt]][[tc]])))

cat("Referential integrity (every key in a result table must exist in its dimension):\n")
print(relationships %>% select(from_table, from_column, to_table, unmatched_keys), n = Inf, width = Inf)
stopifnot("Some keys have no match in their dimension table" = all(relationships$unmatched_keys == 0),
"A dimension key is not unique" = all(map_lgl(unique(relationships$to_table), function(t)
!anyDuplicated(tables[[t]][[relationships$to_column[relationships$to_table == t][1]]]))))
```

In Power BI, relationship autodetection and auto date/time are turned off, and the 35 relationships are built from the list, all many-to-one with single-direction filtering. The measures only look up and format the R results, for example turning three columns into "1.44 (1.34-1.55)". The raw number columns are hidden. Because the results are fixed outputs of the R scripts, the dashboard and this report show the same numbers.

10.2 Report pages

The report has eight main pages, three drill-through pages (group profile, condition detail and risk tier profile) and tooltip pages for the metric glossary, survival curves, hazard ratios, prevalence and causes of death. The theme uses the same rose palette as this report. Every page with slicers has a reset button, and every KPI card has a glossary tooltip that explains what the number means and how it was calculated.

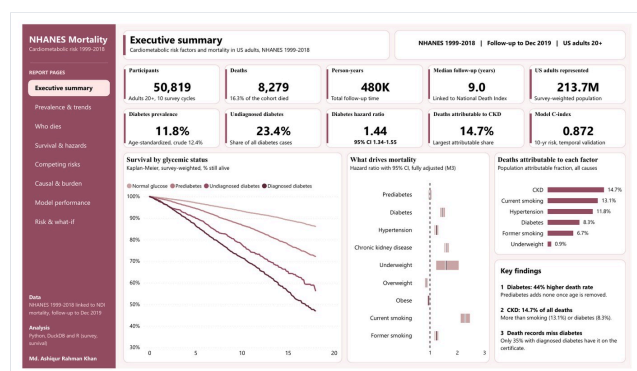


Figure 10.1 Dashboard page 1: Executive summary.

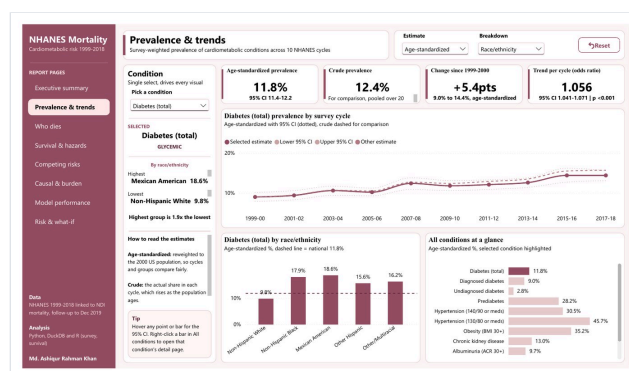


Figure 10.2 Dashboard page 2: Prevalence and trends.

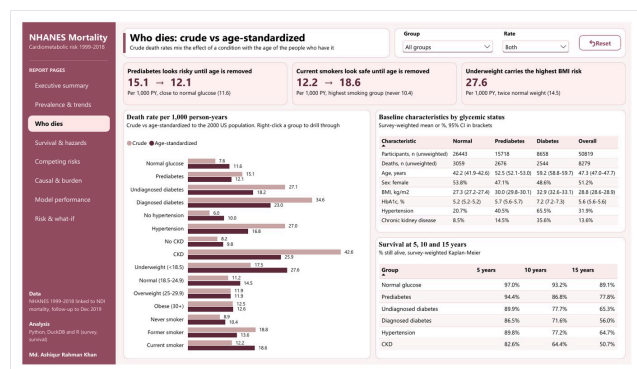


Figure 10.3 Dashboard page 3: Who dies: crude vs age-standardized.

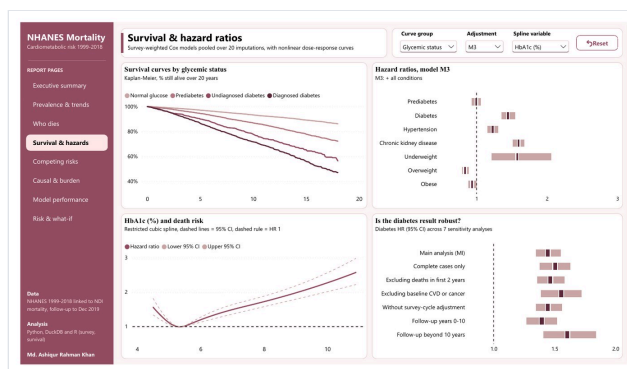


Figure 10.4 Dashboard page 4: Survival and hazard ratios.

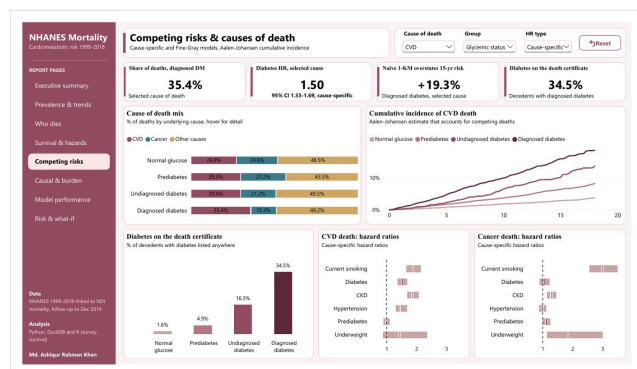


Figure 10.5 Dashboard page 5: Competing risks and causes of death.

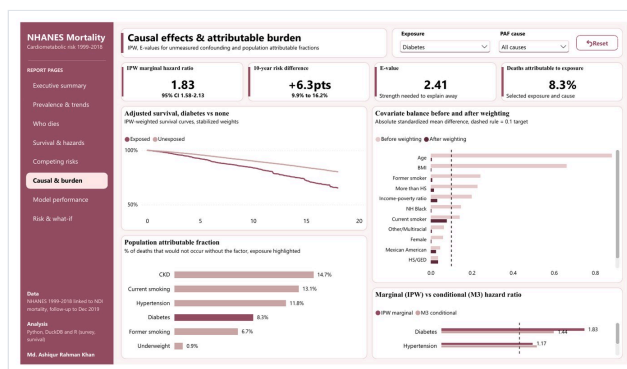


Figure 10.6 Dashboard page 6: Causal effects and attributable burden.

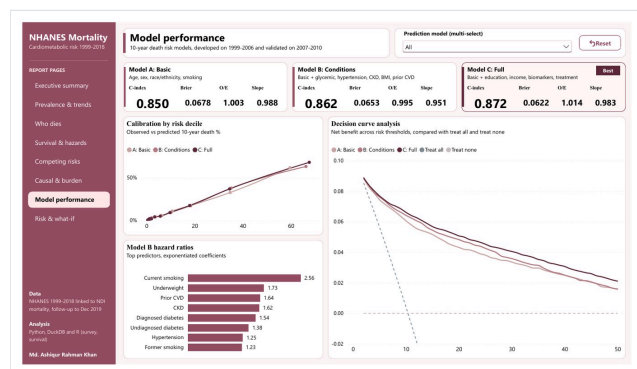


Figure 10.7 Dashboard page 7: Model performance.

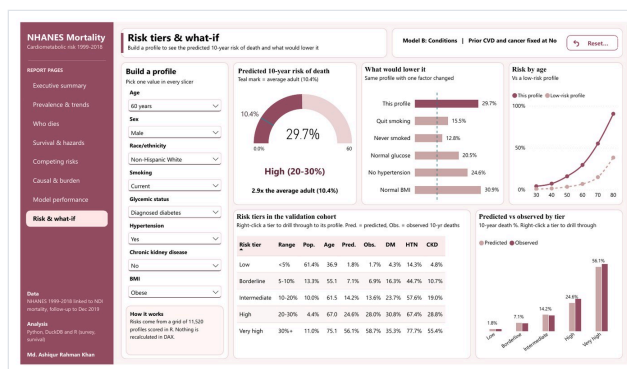


Figure 10.8 Dashboard page 8: Risk tiers and what-if.

11 Discussion and limitations

What the results say together, what they cannot say, and what could be done next.

11.1 Main findings

Age explains more than any condition. Many crude patterns in Sections 3 and 4 changed or disappeared after age adjustment. The clearest case is prediabetes: crude, it looked almost twice as deadly as normal glucose. After adjustment it carried no excess risk in any model (M3 HR 0.99), for any cause of death, in any sensitivity analysis. A prediabetes label on its own says little about risk of death in the next 20 years.

Diabetes and CKD are the strongest cardiometabolic risk factors. Diabetes raised the rate of death by 44% after adjusting for everything else, and by 83% as a total effect that includes its damage to blood pressure and kidneys. Its excess risk is concentrated in CVD and other causes, not cancer. CKD had a hazard ratio of 1.59 and the largest attributable share of deaths, 14.7%. Most of the riskiest profiles in the SQL analysis included CKD.

A quarter of diabetes is undiagnosed, and that has not improved. About 23.4% of adults with diabetes did not know they had it, with no significant trend over 20 years. Undiagnosed diabetes still carried a 43% higher rate of death. Diabetes was also missing from most death certificates of people who had it, so certificate-based counts understate its toll.

Method choices change the numbers. Complete-case analysis would have dropped 22.1% of people who die at almost twice the rate. Treating competing deaths as censoring overstated cause-specific risk by up to 28.5%. Truncating weights, a routine step, made balance worse here. None of these changed the direction of a result, but each changed its size.

Ten-year risk of death can be predicted well from one exam. A model with age, sex, race, smoking, conditions and routine labs reached a C-index of 0.872 on later survey cycles, with close calibration. Most of that comes from age. The conditions and labs add a smaller but consistent gain in ranking, error and net benefit.

11.2 Limitations

- **One measurement at baseline.** Every exposure is measured once, at the exam. People who developed diabetes or CKD during follow-up are counted as unexposed, and treatment changes are not seen. This most likely pulls hazard ratios toward 1, and it may explain part of the change in the CKD hazard ratio after 10 years.
- **Single test for diabetes.** Diabetes and prediabetes rest on one HbA1c or fasting glucose value. Clinical diagnosis needs a repeat test, so some people are misclassified, especially near the cut-offs.
- **HbA1c method change in 2007.** Part of the rise in prediabetes after 2007 is measurement. Survey cycle is in every model and the results do not change without it, but prevalence trends for prediabetes across 2007 should be read with care.
- **Self-reported history.** Prior CVD and cancer are self-reported and may be under-reported.
- **Public-use mortality data.** NCHS perturbs the date and cause of death for some records in the public-use files to protect privacy. This adds some noise to cause-specific estimates but is not expected to bias them.
- **Unmeasured confounding.** Physical activity, diet, alcohol and access to care are not in the models. The E-values show that a hidden confounder would need to be strong to explain away the diabetes and CKD effects, but the hypertension effect is more fragile.
- **BMI paradox.** The lower risk with overweight is probably partly reverse causation from illness-related weight loss, which a single BMI cannot separate.
- **Age top-coding.** Age is capped at 80 or 85, so risk in the oldest adults is estimated with less precision.

- **Prediction scope.** The models were validated on 2007-2010 participants followed to 2019. Their calibration in later years, after changes in treatment and in the population, is not tested.

11.3 Next steps

Three extensions would strengthen the work. Joint models or landmark analysis could use the fact that some exposures change over time. Restricting the prediabetes trend to one HbA1c lab period would separate the measurement effect from a real change. And an external check of model C on another US cohort, or on NHANES cycles once longer follow-up is released, would show whether the calibration holds.

11.4 Conclusion

In US adults followed for up to 20 years, diabetes and chronic kidney disease were the cardiometabolic conditions most strongly linked to death, while prediabetes carried no excess risk once age was taken into account. CKD accounted for the largest share of deaths of any risk factor studied, and about a quarter of diabetes stayed undiagnosed throughout 1999 to 2018. These results depend on using the survey design, handling missing data properly and accounting for competing causes of death. Each of those choices changed the size of at least one estimate in this report.

A Reproducing the analysis

Folder layout, run order and software.

```
NHANES Cardiometabolic Mortality Project/
├── data/
│   ├── raw/           140 NHANES .xpt files, 10 mortality .dat files, download manifest
│   ├── parquet/       one Parquet file per component (15)
│   └── processed/     analytic_cohort.parquet (50,819 rows x 58 columns)
├── notebooks/        01 to 04, Python (Google Colab)
├── r/                05 to 09, plain R scripts (RStudio)
├── outputs/
│   ├── sql_results/   q01 to q11 (notebook 03)
│   ├── tables/        02_* and 04_* tables
│   ├── figures/       04_* figures
│   ├── R/             one folder per R script (05 to 09)
│   └── powerbi/       42 tables, _relationships.csv, _table_dictionary.csv
├── NHANES Analysis Dashboard.pbix
└── NHANES Analysis Dashboard.pdf
```

Run order. Notebooks 01 to 04 run in Google Colab and mount the Drive project folder. Notebook 01 needs internet access to the CDC servers. The R scripts run in order in RStudio, each reading the outputs of the ones before it. Only `root` at the top of each R script needs changing on another computer. Scripts 05 to 07 cache slow steps (the imputation, about 440 Cox fits in script 06, and the Fine-Gray fits in 07) in `.rds` files, so a rerun picks up where it stopped. Deleting a cache file forces a refit.

Randomness. The imputation uses `futuremice` with parallel seed 2026. Everything after it is deterministic given the imputed data.

Table A.1 Software. R package versions from the session info saved by each script.

Software	Version
Python	pandas, NumPy, DuckDB, Matplotlib, PyArrow (Google Colab)
R	4.5.3 (Windows 11)
survey	4.5
mice	3.19.0
mitools	2.7
survival	3.8
tidyverse	2.0.0
arrow	25.0.1
Power BI Desktop	PBIP project, saved as .pbix

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